2024

CALENDAR YEAR REPORT

WASTEWATER TREATMENT FACILITIES

City of Memphis

Division of Public Works

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2024 Calendar Year Report
Wastewater Treatment Facilities
City of Memphis

1.0 Maxson Wastewater Treatment Facility

Introduction-

The contractor performing the Nonconnah Interceptor notified the Maxson Facility on 1-10-2024 they were unable to restart the 96-inch bypass due to heavy grease, 12-18 inches thick and requested the treatment plant increase raw sewage pumping to maintain half-pipe or 44 inches. It snowed 6-inches on 1-15-2024 during the MLK holiday with high temperature of 12 degrees F. Solae had to shut down due to freezing conditions shutting off groundwater wells. Grease began coming over the primary clarifier weirs in large chunks. Temperatures remained sub-freezing for the next few days causing reduction in potable water pressure due to burst pipes. Two main aeration blowers also shut down due to lack of water pressure. Contractor continued having problems installing the bypass plug due to slippery conditions caused by tremendous grease. As the excavation size increased, additional pieces of asphalt and corrugated steel caught in the coarse bar screens. Thick, white greasy foam was observed on the surface of primary clarifiers in January, possibly attributed to Richardson Oil Seed which reported problems with a recirculating pump starting in December. Bioenergy had problems achieving compliance which is common at this time of year. The influent flow meter began reading 15 mgd higher than expected after 1-9-2024 when a new mag meter inverter was installed. Trinova unable to do a true calibration. Maxson Facility water pressure increased to 50 psi on 1-21-2024 allowing for the operation of three main blowers as potable water pressure increased in Memphis. RSP #1 re-installed but unable to get it fully in service for remainder of year due to excessive vibrations, possibly made worse by pump obstruction. Solae reported they were potentially over reporting flow volume by 10% with recent installation of a new mag meter—pipe not flowing full due to lack of smooth transition. This problem would eventually be corrected during the second half of the year. The gravity line from President's Island Harbor Lift Station is expected to be moved back to original location after completion of the ash pond remediation in 2029. It now comes in on the old 60-inch interceptor where it occasionally foams out before connecting to the junction box.

On 2-5-2024, the south coarse bar screen rake malfunctioned due to over torquing as result of debris set loose from surcharging the excavation. This resulted in the need to keep the wet well low until the excavation could be stabilized. On about 2-6-2024, the primary sludge pumping line stopped working which resulted in inability to waste primary sludge for the next several months. On 2-12-2024, the contractor began to install sheet piling to access break to repair 96-inch interceptor. On 2-14-2024, the southern aeration basin side of the plant was temporarily shut down to check out what is going on with Secondary Clarifier 2S inlet valve. This issue would persist for the remainder of the year. Brown foamy grease observed coating the CBS motor and on the primary clarifiers was sampled on 2-23-2024 and sent to Dr. Cliff Lang for analysis. It contained long-chained fatty acids consisting mainly of oleic acid which is of animal and vegetable origin with ¾ non-petroleum-based and remainder petroleum-based. Tetradeoyl Trimethyl ammonium salt (QAC) was also identified which is solid at room temperature and is a cationic surfactant. QAC is an active ingredient in lotions, skin care, and wound treatment. On 2-26-2024, contractor used divers downstream of the bypass plug between break and MH9.

Solae began returning to full production as the year progressed. MIC began adjustments to the influent flow meter to address the problems that started in January.

The Nonconnah Bypass pumping began 3-3-2024 by Xylem having to rely on the Maxson Facility to keep the wet well down to keep the pressure off the plug. A new inlet valve for Secondary Clarifier 2S was ordered as a proactive measure in case it was later determined it needed to be replaced. The wait continued for parts to modify the ARV for primary wasting. Xylem lost a plug on 3-8-2024. The Maxson Facility had three raw sewage pumps (RSP #2, #3, and #4) ready to go as part of this effort to maintain a low wet well. On 3-19-2024, a tremendous amount of detergent-like foam was coming out of aeration basins. Both Maxson Facility and IMPT personnel took their own foam samples on 3-19-2024 and submitted them to Dr. Cliff Lange for analysis. Maxson Facility sample included both underlying mixed-liquor and overlying foam whereas IMPT only sampled the overlying foam. The results showed the FOG in the MLSS to be mainly LCFA with Oleic Acid and Stearic Acid concentrating in Maxson Facility foam at a 11.7 and 8.4 higher rate than in the mixed liquor and stabilized by surfactants. Riviana reported permit excursions for FOG in March. Scott Contractors completed re-piping from ARV and primary pumping put back into service. Starting problems continued with multiple main blowers during the month. Maxson Facility influent flow jumped to 140+ mgd on 3-26-2024 due to rain which realistically was beyond the combined pumping capacity of RSP #2 and #3 and possibly indicative of problems with the influent flow meter. Primary Sludge Pumping was restarted on 3-30-2024 after laying 1000 feet of pipe to Lagoon #1. The bulking secondary clarifier inlet valve stuck at 80% prevented it from being removed from service and contributing to bulking problems. It likely will need to wait until bypass is complete to replace it.

The contractor completed sheet piling on 4-22-2024 and installed dewatering wells. Robert Knecht was on Behind the Headlines PBS TV show on 4-26-2024 stating that HLSD would be disconnected in 8 years (actually, they will just be paying the full City rates if not disconnected). Maxson Facility completed cycling through fine bubble diffuser cleaning dropping DWP from 40-inches to 30-inches using a contractor. Riviana reported another seed oil spill in April.

On 5-6-2024, the Maxson Facility was pumping 155 mgd to 170 mgd due to heavy rains. Again, the reported high pumping rate is suspect due to potential meter over-reporting. New Ashbrook BFPs began arriving using new neat polymer system instead of liquid. Contractor was installing more well points at the Nonconnah Interceptor repair. On 5-8-2024 Valicore was identified as a new central treater discharging to the Maxson Facility with no geographic restrictions from where it can receive waste. TF 1 arm tore up on 5-11-2024. By 5-14-2024, almost all the pipe was exposed on the Nonconnah Interceptor. On 5-24-2024, the Maxson Facility received an NOV from TDEC for QNC for February, March, and April. On 5-31-2024, the 96-inch plug was caught in the CBS. Contractor began repair of #2 primary clarifier floor and center drive replacement during the month. TF5 distribution arm twisted and fell off during the month. Assessment of the 60-inch bypass line was completed and determined to be ok based on water level comparison. Only concern is where pipe discharge separated from headwall. US Peroxide personnel visited Solae lift station and discharge point of force main into the Nonconnah Interceptor. TF1 distribution arm twisted and fell off about this time frame. RSP #1 momentarily returned to service and then was removed from service due to excessive vibration. Based on feed/spray dryer reporting, Solae appeared to be at full production at this point but at 2/3's historical discharge loading.

On 6-2-2024, local news media reported that xAI was building the world's largest supercomputer at the former Electrolux site. This would involve accelerating construction of a grey water facility to deliver cooling water to xAI, TVA, City of Memphis, and future commercial and industrial users. On 6-7-2024, the Mayor announced construction of effluent water treatment facility which would provide grey water to aXI, TVA, City of Memphis, and future commercial and industrial users reducing the demand for aquifer water. The Mayor had brought up the grey water solution. MLGW had been working on a gray water solution (as related to TVA Combined Cycle) but felt it was several years away. CDM Smith was engaged by the City to evaluate influent, primary effluent and secondary effluent before disinfection for tertiary treatment employing Membrane Bioreactors. It is noted that TVA previously had issues with Total Dissolved Solids when considering Maxson Facility effluent for tertiary treatment. The Maxson Facility would continue to plan to use potable water for odor control instead although evaluation on a strainer system was continuing. Meanwhile, CDM Smith began design of a solar farm on land previously planned for wind turbines. The Maxson Facility found where the 96-inch plug wrapped around North CBS on the second floor after coming loose from bypass. On 6-21-2024, the plug was removed and the rake had been damaged. On 6-22-2024, news media reported that xAI brought in 450,000 gallons of potable water by truck to start the closed-loop cooling system. On 6-22-2024 the Maxson Facility begam producing enough biogas to start drawing biogas from covered lagoons. On 6-24-2024 the Maxson Facility was only able to operate TF3 with TF1, 2, and 4 no longer working due to distribution arm failure being part of the original second set of distribution arms for TF1-4. By 6-24-2024, Maxson Facility installed the spare CBS rake xAI submitted a discharge permit application to discharge net 1.2-1.3 mgd after 8 loops in cooling water recycle. xAI will be discharging in the same manhole formerly used by Electrolux which happens to be the overflow manhole used by the Maxson Facility during wet well surcharge conditions. The City continued negotiating a consent order with Solae to reduce spent flakes by 80% over a 2-year period. The Maxson Facility was able to run 3 main blowers since 6-21-2024 which appears to be the limit with FBD although there were still motor protection relay concerns.

On 7-8-2024, the Maxson Facility still has only TF with a functioning distribution arm. The State of Tennessee indicated they would close the park road so the collapsed interceptor would not need to be replaced within an encasement. However, 17 other road crossings have the same potential for failure in the near to long-term, maybe longer term based on lessening surcharge conditions encountered further from the plant. The Maxson Facility EDS began working on about 7-9-2024, ushering in a new era of data sharing, although bugs needed to be worked out. The 60% deliverables for the solar farm were announced as the City shifted from wind to solar while ignoring the less sexy and problematic biogas. A conceptual design report for the grey water facility using MBR technology was prepared by CDM Smith. City leaders met with xAI and MLGW on 7-30-2024 to review the competing conceptual designs for the grey water plant offered by CDM Smith and Gresham Smith as the City and xAI played dueling consultants. CDM reduced the cost for the MBR plant to $170 million while aAI recommended the $90 million ceramic filtration system proffered by xAi. Potential grey water customers include xAI, TVA, Nucor, and the City. In both scenarios, ammonia will need to be removed. It is noted that ammonia in Maxson Facility influent ranges up to 30,000 ppd with the majority likely produced by the lagoons as only 500-5000 ppd reported in the SMRs. By the end of the month, 60 feet of Nonconnah Interceptor piping has been laid leaving 310 feet to go.

In August, xAI asked how much money it would cost to construct an entirely new Maxson Facility from scratch to provide the required tertiary treatment. Having chosen Gresham Smith as their design engineer, xAI changed their conceptual design based on regular MBR technology to be sized to handle the needs of TVA, Nucor, and xAI with the City electing not to participate. The design would involve denitrification which would require a carbon source. The Maxson Facility was still unable to waste primary sludge due to Lagoon 2A project not being quite finished resulting in the build-up of floatables in the primary clarifiers. On 8-9-2024, Kellogg's announced it was cutting back on production due to nation-wide restructuring of the supply chain. The last joint of pipe for Nonconnah Interceptor repair was laid on 8-23-2024. MIC (Mike Durham) got most of basin meters working as the Maxson Facility continued putting more emphasis on flow meter validation. xAI indicated that their MBR design would discharge 1% biological sludge at 0.5 mgd to Maxson Facility lagoon for digestion. The treated effluent would go to xAI, TVA and Nucor with the remaining treated effluent being discharged to the sewer system. Maxson Facility effluent would be utilized after chlorine disinfection in accordance with TDEC regulations for water re-use. xAI announced the construction if a second data center located in the Whitehaven area of Memphis on the Mississippi State line. Solae continued to be hiring new people as it recovers from last year's strike so unable to ascertain their expected loading. Evaluation of strainer technology continued and was expected to produce water for general plant use but not for main aeration blowers or raw sewage pumps. Grey water plant was expected to be up and running in 1-1 ½ years (we will see).

During September, as the strainer evaluation continued, it was opined that the Maxson Facility could use the strained effluent for pump seal water as well as odor control media water. However, the Maxson Facility would continue to use potable water for raw sewage pump and aeration blower cooling until it was feasible to use strainers. On 9-4-2024, the Nonconnah Interceptor Bypass was shut off and plugs removed. xAI released their Grok Artificial Intelligence system on about 9-2-2024 for public use after building the H100 training cluster in 122 days with plans to double in size in the coming months. No. 2 RAS blower motor was put into service after repairs made due to bad vibrations. A Consent Order with Solae was enacted regarding 40,000-60,000 ppd of spent flakes discharge to be reduced by 50% and then 80% in 3 years. On 9-23-2024, it was announced that xAI would purchase 15-acres of Maxson Facility property for construction of the grey water facility. This is the property the City purchased 12 years ago for nutrient removal. xAI will be wasting into Lagoons 2A and 2C as backup with a 4-inch diameter wasting line. On 9-27-2024, US Peroxide conducted a field survey of H2S levels noting 4 pH where Solae force main discharged into the Nonconnah Interceptor. It may be impossible to control since pH stays below 6 and iron does not bind to H2S in acidic conditions. A data logger located at the Maxson Facility headworks showed 100-400 ppm H2S. Large scale collection system evaluation is apparently required. Caustic shocking at Solae would be recommended and 50% peroxide addition upstream of head works. Approximately 100-150 gpd peroxide solution would be recommended to reduce H2S at headworks by 50%. Influent flow increased to 150 mgd on 9-30-2024 due to rain. The Maxson Facility limped along unable to waste primary sludge and limited waste activated sludge due to partially plugged discharge line. RSP #1 appeared to need new discharge valve after motor rewind and pump rebuild performed. Chris Hill began mobilizing for the primary sludge pumping upgrade project while doing work on other plant projects. The grit crane went out of service for the umpteenth time, this time due to having problems with the trolley. Therefore, a pump was used to remove grit. Solae effluent flow meter appeared to be correctly reading now allowing for more accurate SMR reporting and evaluation of impact on the Maxson Facility.

On 10-4-2024, xAI met with Edge and announced the lease of 552 acres of adjoining vacant land, east of Nucor, for various possible uses. The lease was for 21 years with 2 10-year options with the first year free. The land apparently was formerly leased by a farmer in the 1980s and used for land application of biosolids as part of the Earth Complex. The grey water facility was christened the Colossus Water Recycle Facility (CWRF) with three formal project partners consisting of xAI, TVA, and Nucor and designed for 13 mgd based on future expansion. A service agreement with Hach for the recently acquired D.O. probes was initiated. On 10-17-2024, the Stiles Facility dredge was moved to the Maxson Facility to assist with sludge removal from Lagoon 5. The grit tanks were returned to service Solae began removing spent flakes targeting 40,000 ppd for removal with a goal of 80% removal. However, no impact was apparent on influent BOD.

On 11-8-2024, final connection to the Maxson Facility EDS located at the Main Library was made after extensive trouble shooting. During recent rain, it was observed that the Maxson Facility can only get 80 mgd through half the plant during half plant operation. A kick-off meeting was held on 11-14-2024 for the interceptor sampling project. On 11-19-2024, it was observed that Maxson Facility disinfection graphics in Ovation originally set up by CDM Smith were based on the Stiles Facility and never updated when the disinfection changed to gravity and COD. A strong petroleum odor was detected at the Maxson Facility on 11-14-2024. Surveyors were observed surveying boundaries for the proper acquisition for the future Colossus WRF. Chris Hill Construction personnel installed both the sweeper arms on Primary Clarifier #2 and a conveyor belt on the new belt filter presses.

Maxson Facility met with US Peroxide personnel on 12-3-2024 regarding locations near the Maxson Facility for sulfide control. The pH was too low to add iron salts at the Solae lift station. Again, the recommendation was made to add caustic to raise pH and kill bacteria at the Solae lift staton. On 12-5-2024, it was discovered that disinfection points were never added to the Maxson Facility EDS because you physically need to be at the EDS to add the points. On 12-23-2024, xAI received approval from the TVA Board to go up to 150 MW at the Colossus Facility.

The Maintenance Department completed numerous noteworthy tasks during the year. In January, a 4-inch water line was set up and run from the Service Building to Manhole #1 on the 60-inch bypass line. The # 1 Primary Sludge Pump was completely rebuilt with new rotor, stater, seals, and bearings installed. The #1 Raw Sewage Pump was removed and pump, mag drive, and motor rebuilt and reinstalled.

In February, #3 North Return Sludge Pump was disassembled and new expansion joint installed. At #3 Screw Pump, a new gearbox was purchased and installed and the upper rotating unit was refurbished and installed. #2 North Return Sludge Pump was disassembled and new expansion joint installed. The Borger Grease Pump was rebuilt and reinstalled.

In March, a new vacuum pod was installed on #2 North Return Sludge Pump. #7 Screw Pump was refurbished, and new upper rotating unit installed. At #3 South Blower, platform oil lines were replaced, joints welded, and flange connectors welded on. Two screw pump upper rotating unit housings were fabricated for future use. # 2 Primary Clarifier was drained and cleaned, stoppage from underflow drain line was removed, and put back in service. A 12-inch by-pass line was set up from Manhole #7 to Lagoon #1 to bypass the 16-inch primary wasting line.

In April, #2 Fine Bar Screen Assembly screen rake was completely rebuilt and new cog wheels, bearings, grease assembly and guide arms installed. # 1 Effluent Water Pump was rebuilt and put back into service.

In May, #1 Primary Sludge Pump was completely rebuilt and new rotor, stater, seals, bearings installed. A new connector coupling was fabricated and installed for # 6 Primary Sludge Pump from motor to pump. A 10' x 18' x 12" concrete pad and canopy for open centralized oil staging area was built for the Lagoon.

In June, the 12-inch valve on #5 Lagoon going to 110 South was replaced. At the Aeration Basins, a new air header was fabricated and 2-inch temporary water lines run to all blowers so that the permanent water line could be replaced. At the Blower Building Electrical Room, the 15-ton Trane HVAC unit was removed and replaced with a new unit for the electrical room.

In July, a new 18-inch steel flange and 12-foot long 18-inch stainless steel chute was fabricated for the press and installed at #2 Fine Bar Screen. Lagoon Monitoring Well # 7 was removed from service and new well installed in the same vicinity. New epoxy coating was placed on 48", 30" 24" and 12" piping throughout Pipe Gallery. For the Effluent Water Filter System, new filter cannisters, four new 6-inch butterfly and eight new 8-inch butterfly valves installed, and the header was primed and painted.

In August, a new set of Gould shafts for future pump rebuild was fabricated for the ABF Pumps. At #1 South Blower, the platform oil lines was replaced, joints welded, and flange connectors welded on. A compressor cover was retrofitted for new switch gear installation. The inoperative drive was removed and new center drive gear reducer installed on #2 Primary Clarifier. Broken weirs were repaired and/or replaced on # 3 Primary Clarifier.

In September, a new shaft was fabricated, new bearings and seals installed, and pump rebuilt and installed at #2 South Return Sludge. At Primary Clarifier #2, the defective center drive and center column was replaced, catwalk rehabbed, sweep arm structure leveled/aligned, and concrete floor regrouted. At #3 ABF Pump, an actuator was removed, refurbished, and installed on the 36-inch discharge line. Broken weirs were repaired and/or replaced on # 2 Primary Clarifier. A new 24-inch swing check valve was purchased and installed on pump discharge line at #3 North Return Sludge Pump. A new gear box was purchased and installed and upper rotating unit refurbished and installed at #1 Screw Pump.

In October, defective air relief valve # 2 was replaced and upgraded from brass to all stainless steel material on the 16-inch Primary Sludge line. At #3 Screw Pump, the motor base was removed, reconfigured, and refurbished motor base to accommodate new motor base reinstalled with new motor. Broken weirs were repaired and/or replaced on #1 Primary Clarifier. The electrical cable tray was replaced on North Coarse Bar Screen. Aeration Blower Control Room HVAC was completely rehabbed, and condenser coil and fan relocated from the Blower Building roof to the road level for ease of maintenance.

In November, a leak on the west side of Lagoon 2C Building was repaired. The 1-1/2-inch discharge pipe from manhole north of Lagoon 2A was repaired. The Aeration Blower Air Filter Room door was replaced to better seal and lock tight the filter room. The leaking 18-inch ductile

iron pipe for Lagoon 2C pump discharge pipe was excavated and replaced with HDPE pipe. At Lagoon # 1, eight defective stationary centrifugal rainwater pump units mounted at the lagoon banks were replaced with submersible pumps mounted on top of the covers and all electrical components rebuilt. Twenty total holes on Lagoon 1,2B, & 2C covers were patched and repaired. The area around Manhole #HL010150 near the Horn Lake Metering Station was built up for flow study. The old fence south of the new Administration Building was removed.

In December, Phase 1 of the new security perimeter fencing east of the new Administration Building was completed.

In addition to the major tasks listed above, the Maintenance Department completed 5713 Preventative Maintenance Work Orders, 13 Scheduled Maintenance Work Orders, and 313 Unscheduled Maintenance Work Orders for a total of 6039 work orders. The annual number of work orders completed over the last 10 years is summarized as follows:

Year	2024	2023	2022	2021	2020	2019	2018	2017	2016	2015
Preventative	5713	5733	4967	5067	7435	3514	3417	1896	178	567
Scheduled	13	242	175	558	1153	871	799	321	20	72
Unscheduled	313	298	304	622	1083	1756	1190	252	18	118
Total	6039	6273	5446	6247	9671	6141	5416	2469	265	772

Ferrous Chloride Use –No ferrous chloride has been used at the Maxson Facility since January 2012. Among its varied uses, ferrous chloride reduces the likelihood of struvite formation by producing acidic conditions and binding up magnesium and phosphorus ions. Struvite formation began to be observed in Lagoon Nos. 3 and 4 in 2015, favoring areas where there is turbulence. A 5000-gallon tank has been placed into position to add ferrous chloride to feed sludge prior to dewatering when this practice is resumed. No ferrous chloride was purchased or used in 2024. In the future, purchase options would include commercially made ferrous chloride containing about 34% or spent pickle liquor containing 10% iron.

Flow – The average daily flow rate in 2024 was 76 mgd which was 10 mgd higher than the previous year. The 0.5^{th} percentile flow in 2024 was 52 mgd which is considered base flow for the year. The average daily flow received from the major significant industrial users was 10 mgd. This leaves 14 mgd for 82% of average daily flow accounted for. The total rainfall for the year was 59 inches (NWS). Memphis annual rainfall has averaged 54 inches.

BOD_5 – The average influent BOD_5 loading for 2024 was 438,744 ppd day, which is 50,262 pounds or 11% increase from the previous year. The average influent BOD_5 concentration was 704 mg/l and the average effluent concentration was 129 mg/l, resulting in a removal efficiency of 82%. The total average BOD_5 reported being discharged by the major significant industrial users was 260,963 ppd in 2024, equivalent to 59% of the total loading. This is 41,838 pounds or 16% more in monitored industrial loading from the previous year. The 0.5 percentile loading received in 2024 was 119,052 ppd which is the assumed domestic loading for this year. This leaves 58,729 ppd of influent BOD_5 for 87% of average influent BOD loading accounted for.

Total Suspended Solids (TSS) – The influent TSS averaged 389,204 ppd, which is 57,364 pounds or a 15% increase from the previous year. The average influent TSS concentration was

613 mg/l and the average effluent concentration was 123 mg/l resulting in a removal efficiency of 80%. The total average TSS reported discharged by the major significant industrial users was 153,002 ppd in 2024, equivalent to 39% of the total loading. This is 28,522 pounds or 19% increase in monitored industrial loading from the previous year. The 3 percentile loading for 2024 was 97,367 ppd which is expected domestic loading for this year. Lagoon supernatant and dewatering filtrate return was not sampled and is believed to be negligible. This leaves 138,835 ppd of influent TSS for 64% of average influent TSS loading accounted for.

Chemical Oxygen Demand (COD) - The influent averaged 1566 mg/l and 975,670 ppd, which is 148,309 pounds or 15% increase from the previous year. The effluent averaged 411 mg/l and 260,508 ppd. The average removal was 74%. The 2 percentile loading for 2024 was 278,645 ppd which is expected domestic loading for this year. The total average COD discharged by the major significant industrial users was 483,889 ppd which equals 50% of the total loading. This leaves 213,136 ppd for 78% of influent COD accounted for. The COD surcharge is based on covering 50% of disinfection costs.

Effluent E. coli - E. coli analysis was performed on daily effluent grab samples throughout the year as required by the NPDES permit as report only. Effluent E. coli results for this past year are presented in Figure 1 showing the periods of initial and final startup of disinfection.

Water – In 2024 metered water usage was 32,207 CCF per month resulting in total usage of 386,478 CCF or 0.8 MGD. The total annual cost was $506,490. This is a decrease of 23,816 CCF per month from the previous year. The water meter was out of service beginning August 2024.

Energy – The main power consumption of the plant was an average of 3,939, 276 kwh per month in 2024. This is 25,831 less per month than the previous year. The monthly peak demand averaged 6916 kW which is 510 kW more than the previous year. The average monthly main power bill was $392,980, which is $57,069 more than the previous year. Payments for participating in the ENERNOC curtailment program for main power are unavailable but are likely minimal due to construction activity and equipment problems.

April was the high month for demand at 9746 kw and the high month for consumption at 4,574,361 kwh. October was the low month for demand at 5332 kw and the low month for consumption at 3,234,932 kwh. Daily electrical demand throughout 2024 is shown in Figure 2.

Effective 10-1-2015, the City entered into an agreement with MLGW to switch from Seasonal Demand to Time of Use billing. TVA would charge Fuel Cost Adjustment based on user class. TOU rate structure benefits high load factor customers through incentivizing off-peak energy usage. The contract term is 5 years and will be automatically extended unless otherwise terminated.

Sludge and Biosolids – The quantity of primary sludge wasted to the lagoons in 2024 was 42.1 million pounds, a decrease of 7.7 million pounds from the previous year. The quantity of secondary sludge wasted to the lagoons in 2024 was 62.5 million pounds, a increase of 3.8 million pounds from the previous year. The total quantity of both primary and secondary sludge wasted from the plant was 104.6 million pounds, a decrease of 2.9 million pounds from the previous year.

Approximately zero million pounds of digested sludge was dewatered at the secondary dewatering facility and disposed in the 800 acres available for Surface Disposal, of which 309 acres is considered permanently disposed. This includes approximately zero million pounds of iron from previous years that was added to Lagoon Nos. 3, 4, and 5 (the uncovered lagoons). Approximately zero % of the combined primary and secondary sludge wasted to the lagoons in 2024 was destroyed equivalent to zero million pounds. The formula used to calculate percent destroyed is %VS of wasted primary and activated sludge multiplied by %VAR. The %VS for primary and waste activated sludge is based on two separate sampling points and then combined mathematically based on proportion. The %VS for feed sludge is based on a single sampling point. Note that it was not possible to calculate the amount of sludge destroyed since no dewatering was performed during the year.

Considering removal and volatilization the new change in the lagoon solids inventory was an increase of zero million dry pounds within the approximately 257 acres of covered and open lagoons. The calculated cumulative dry solids within the covered and uncovered lagoons is 766,802,224 million dry pounds. The calculated dry bio-solids that has been surface disposed since 1979 is 1,527,402,725 million dry pounds.

A total of zero million pounds of digested sludge was disposed in 2024 as surface disposed biosolids (zero tons) and volatilized lagoon sludge (zero tons) at an estimated total cost of $4,652,317.

The annual cost per dry ton for digestion, dewatering, surface disposal and volatilization for the past 10 years is as follows:

2024	2023	2022	2021	2020	2019	2018	2017	2016	2015
$89	$344	$98	$108	$74	$112	$67	$66	$76	$68

The annual cost per dry ton for digestion, dewatering and surface disposal for the past 10 years is as follows:

2024	2023	2022	2021	2020	2019	2018	2017	2016	2015
NA	$3,481	$406	$468	$291	$549	$265	$222	$201	$214

The annual amount of digested sludge (in million dry pounds) removed from the lagoons over the last 10 years is provided below:

2024	2023	2022	2021	2020	2019	2018	2017	2016	2015
0	2.8	21.7	20.4	30.7	15.8	23	24.7	29.95	27.2

For the last 10 years, net accumulation has been occurring at an average rate of 50.5 million dry pounds per year. Surface disposal has averaged 19.6 million dry pounds per year. To stop net accumulation, the Maxson Facility will need to de-water and surface dispose an average of 70.1 million dry pounds per year which is 3.6 times the existing rate and would be 97 percentile level of performance. Hopefully, the upcoming project to clean out, subdivide, and cover Lagoon 5 will correct this situation.

Biogas – For the year 2024, the Maxson Facility produced a total of zero million cubic feet of biogas. TVA used zero million cubic feet of the annual production.

The annual biogas produced by the Maxson Facility for the past 10 years is as follows (in millions of cubic feet):

2024	2023	2022	2021	2020	2019	2018	2017	2016	2015
<1	<1	191	430	438	552	565	744	1167	1254

The annual daily average biogas produced for the past 10 years is as follows (in millions of cubic feet):

2024	2023	2022	2021	2020	2019	2018	2017	2016	2015
<1	<1	0.52	1.2	1.2	1.51	1.55	2.04	3.20	3.44

Annual biogas yield for the last 10 years (ratio of biogas to volatiles destroyed) in cubic feet per pound is as follows:

2024	2023	2022	2021	2020	2019	2018	2017	2016	2015
NA	0.04	2.8	6.3	4.9	8.9	8.3	12.4	23.8	21.4

The amount of sludge volatilized each year is relatively constant with no discernable long-term trends or impact on the volume of biogas produced. What is variable is the amount of biogas produced per pound of sludge destroyed as shown above. Since 2017, biogas volume has dropped to levels not seen since 2003 before Lagoons 2A, 2B, and 2C were first covered. Trends in biogas production over the years are shown in Figure 3.

SCHD issued the final Title V permit on 10-4-2019, Permit # 00701-01TV, expiring 10-4-2024. A new permit application would be required by 4-27-2024.

The August-December 2022 Semi-Annual Report was transmitted to the SCHD in February. SCHD performed annual inspection on 3-23-2023. The January-July 2023 Semi-Annual Report was transmitted to the SCHD in August.

The annual biogas usage by TVA along with the amount of biogas flared as waste gas by the Maxson Facility over the last 10 years is as follows (in millions of cubic feet):

Year	2024	2023	2022	2021	2020	2019	2018	2017	2016	2015
TVA	0	0	0	0	0	0	0	0	0	0
Flared	0	1	191	430	438	552	565	744	1167	1254

The City received zero in revenue from TVA for the Maxson biogas for 2024.

Permit Violations – This year the Maxson Facility had 812 excursions of its NPDES permit limits related to treatment efficiency and performance. The number of excursions per month was steady throughout the year and a function of the number of trickling filters in service, ability of the blowers to meet air demand, and changes in influent loading due to industrial activity. There

were 1-3 trickling filters and 2-3 blowers in service each month and effluent quality reflected this. The number of excursions increased or decreased due to extenuating circumstances as described as follows:

In January, there were 122 permit excursions which was an increase from 95 the previous month. There were BOD monthly average concentration, monthly average pounds, monthly average percent removal, 4 weekly average concentration, 4 weekly pounds, 23 daily concentration, and 5 daily percent removal excursions. There were TSS monthly average concentration, monthly average pounds, monthly average percent removal, 4 weekly average concentration, 4 weekly average pounds, 25 daily concentration, and 19 daily percent removal excursions. There were 13 effluent dissolved oxygen and 15 settleable solids excursions. The increase in violations was due to reduced ability to waste both primary and secondary sludge beginning late 2023. This was related to the temporary relocation of primary and secondary sludge discharges in mid-2022 away from Lagoon 2A to the south side of Lagoon 1 and resulting problems.

In February, there were 111 permit excursions which was a decrease from 122 the previous month. There were BOD monthly average concentration, monthly average pounds, monthly average percent removal, 4 weekly average concentration, 4 weekly average pounds, 29 daily concentration, and 3 daily percent removal excursions. There were TSS monthly average concentration, monthly average pounds, monthly average percent removal, 4 weekly average concentration, 3 weekly average pounds, 17 daily concentration, and 9 daily percent removal excursions. There were 28 effluent dissolved oxygen and 4 settleable solids excursions. The number of violations was reduced when secondary sludge wasting returned to normal after the secondary sludge discharge outlet was relocated.

In March, there were 89 permit excursions which is a decrease from 111 the previous month. There were BOD monthly average concentration, monthly average pounds, monthly average percent removal, 5 weekly average concentration, 5 weekly average pounds, 25 daily concentration, and 1 daily percent removal excursions. There were TSS monthly average concentration, monthly average pounds, monthly average percent removal, 4 weekly average concentration, 4 weekly average pounds, 17 daily concentration, and 2 daily percent removal excursions. There were 1 effluent settleable solids and 19 effluent dissolved oxygen excursions. The number of violations continued to be reduced as the plant returned to normal wasting and the primary sludge discharge outlet was relocated toward the end of the month.

In April, there were 68 permit excursions which is a decrease from 89 the previous month. There were BOD monthly average concentration, monthly average pounds, 4 weekly average concentration, 4 weekly average pounds, and 23 daily concentration excursions. There were TSS monthly average concentration, monthly average pounds, 4 weekly average concentration, 4 weekly average pounds, 15 daily concentration, and 1 daily percent removal excursions. There were 1 settleable solids and 8 dissolved oxygen daily effluent excursions. The number of violations was reduced with the return to normal wasting.

In May, there were 55 permit excursions which is a decrease from 68 the previous month. There were BOD monthly average concentration, monthly average pounds, 4 weekly average concentration, 3 weekly average pounds, and 10 daily concentration excursions. There were TSS monthly average concentration, monthly average pounds, monthly average percent removal, 4 weekly average concentration, 4 weekly average pounds, 11 daily concentration, and 2 daily

percent removal excursions. There were 8 settleable solids, and 4 dissolved oxygen excursions. The number of violations was reduced with the return to normal wasting.

In June, there were 51 permit excursions which is a decrease from 55 the previous month. There were BOD monthly average concentration, monthly average pounds, 4 weekly average concentration, 3 weekly average pounds, and 12 daily concentration excursions. There were TSS monthly average concentration, monthly average pounds, 5 weekly average concentration, 5 weekly average pounds, and 14 daily concentration excursions. There were 1 settleable solids, and 3 dissolved oxygen excursions. The number of violations was reduced with the return to normal wasting.

In July, there were 80 permit excursions which is an increase from 51 the previous month. There were BOD monthly average concentration, monthly average pounds, monthly average percent removal, 4 weekly average concentration, 4 weekly average pounds, and 30 daily concentration excursions. There were TSS monthly average concentration, monthly average pounds, 4 weekly average concentration, 4 weekly average pounds, and 27 daily concentration excursions. There were 1 settleable solids, and 1 dissolved oxygen. The number of violations increased due to a malfunctioning primary clarifier.

In August, there were 49 permit excursions which is a decrease from 80 the previous month. There were BOD monthly average concentration, monthly average pounds, monthly average percent removal, 5 weekly average concentration, 5 weekly average pounds, and 24 daily concentration excursions. There were TSS monthly average concentration, monthly average pounds, 3 weekly average concentration, and 6 daily concentration excursions. There was 1 effluent settleable solids excursion. The number of violations was reduced with new TF 5 going into service.

In September, there were 41 permit excursions which is a decrease from 49 the previous month. There were BOD monthly average concentration, monthly average pounds, monthly average percent removal, 3 weekly average concentration, 2 weekly average pounds, and 17 daily concentration excursions. There were TSS monthly average concentration, monthly average pounds, monthly average percent removal, 2 weekly average concentration, 1 weekly average pounds, 8 daily concentration excursions and 1 daily percent removal excursions. There was 1 effluent dissolved oxygen excursion. The number of violations was reduced with new TF 6 going into service.

In October, there were 37 permit excursions which is a decrease from 41 the previous month before. There were BOD monthly average concentration, monthly average pounds, monthly average percent removal, 4 weekly average concentration, 2 weekly average pounds, 19 daily concentration, and 2 daily percent removal excursions. There were TSS monthly average concentration, monthly average pounds, 1 weekly average concentration, and 4 daily concentration excursions. The number of violations was reduced with the new RAS reaeration tank going into service.

In November, there were 51 permit excursions which is an increase from 37 the previous month. There were BOD monthly average concentration, monthly average pounds, monthly average percent removal, 4 weekly average concentration, 4 weekly average pounds, and 18 daily concentration excursions. There were TSS monthly average concentration, monthly average

pounds, monthly average percent removal, 4 weekly average concentration, 3 weekly average pounds, 8 daily concentration, and 2 daily percent removal excursions. There were 1 effluent settleable solids and 1 effluent dissolved oxygen excursions. There number of violations increased due to south side aeration being out of service.

In December, there were 58 permit excursions which is an increase from 51 the previous month before. There were BOD monthly average concentration, monthly average pounds, monthly average percent removal, 4 weekly average concentration, 3 weekly average pounds, 18 daily concentration, and 1 daily percent removal excursions. There were TSS monthly average concentration, monthly average pounds, monthly average percent removal, 4 weekly average concentration, 3 weekly average pounds, 11 daily concentration, and 5 daily percent removal excursions. There were 3 daily settleable solids effluent excursions. There number of violations increased due to south side aeration being out of service.

The cumulative violations are shown in Figure 4 with Stiles Facility cumulative violations for comparison. The excursions associated with different major periods of plant history are summarized below:

Period	Average Number of Excursions Per Year	Description
1982-1985	30.5	First plant capacity expansion (Consent Decree)
1986-1994	4.0	Industrial loading increasing toward full capacity
1995-1997	24	2nd plant capacity expansion (Voluntary)
1998-2010	4.4	Steady loading conditions
2011-2020	51.5	Collection system and plant equipment failures; subsequent plant equipment replacement projects
2021-2024	908	3rd plant capacity expansion (Voluntary)

The Maxson Facility has had two years (1987 and 2010) with zero violations. The highest number was 1524 violations in the year 2022.

During 2011-2024, 456 of 4146 violations at the Maxson Facility, or 11%, were for “pounds” permit limit excursions which the Stiles Plant does not have to report, creating regulatory imbalance between the treatment plants.

Disinfection, Process Improvements, and Odor Control

In 2014, the disinfection, process capacity and treatment improvements, and odor control projects were combined into a single initiative called Construction Manager at Risk (CMAR). Treatment capacity increase was based on an average daily flow of 90 mgd at influent BOD

concentration of 800 mg/l equivalent to 600,000 ppd as well as compliance with 30-30 BOD and TSS performance standards. A new final completion date of 3-8-2027 was announced for the CMAR.

Package 2A / GMP 3 covering construction of two new secondary clarifiers, intermediate pumping improvements, odor control, and fine bubble diffusers continued from the previous year. By early year, everything had been completed except the odor control system. In February, final walk-thru was being scheduled for clarifiers and RAS pumping station. By the time of the 5-16-2024 CMAR update, it was reported that the contractor had finished installing replacement gaskets and hardware on south and north aeration basins headers and started up odor control. In May, odor control was not complete due to wiring problem. On 6-18-2024, BioAir was on-site to complete installation and begin startup of the odor control system after the odor control transformer replacement arrived. Work continued installing flow control structure gates.

In July, work on GMP 3 punch list items continued. Problems continued with the start-up of the BioAir system H2S levels to be treated by the BioAir system were significantly higher than anticipated compared to the original sampling done 8 years ago with severe H2S levels at the Screw Pumps and FBS. In September, Bio-air system was still being tweaked. There were pumping problems with the new secondary clarifiers. In October, work continued installing flow control structure gates. North side aeration was shut down on 10-28-2024 so CDM Smith could install a flow control structure. In October, odor control system electronics problems continued. In October waiting continued on flow control gates. In November, BioAir system began being seeded with microbes and nutrients. Southside aeration was still out of service as CDM Smith worked on the splitter box. December began with the Maxson Facility continuing to run on the north aeration system due to RAS splitter box gates being installed. December was spent waiting on biomass to form in the BioAir system.

Package 2B / GMP 4 for constructing re-aeration tanks and a new blower building continued from the previous year. By early year, waiting continued for relocation of the RAS oil cooler. In April, installation of all FBD in the RAS tank was completed and 2 of 3 RAS blowers were working. In May, the connection between the two new clarifiers and the RAS tank was made. In June, efforts continued to get the 3rd RAS blower running due to needing new coupling. In September, RAS Blower #2 was installed after repairs. In October, test runs were started on the new RAS blowers which are air-cooled. In November, oil leaks on the RAS Blowers were repaired and final tie-ins to RAS tank were made. GMP 4 critical path was the transformer. In December, final walk thru was performed and work on the punch list continued.

Package 3 / GMP 5 for constructing and installing new bio-tower media, odor dispersion fans, and tower improvements continued from the previous year. In January, the wait on coating TF 4, 5, and 6 began. Another batch of media began to arrive. In February, TF 5 and 6 troughs were cleaned up. A new flowline for TF5 trough was poured and cured. Crack repair began on TF4. In April, TF5 and TF6 trough coating testing began after sloughing off. This occurrence called into question trickling filter coating applied project to date. Weirs were installed on the troughs to catch debris. On 5-20-2024 media removed from TF 4, 5, and 6 were taken to the nearby Atlas Facility, previously the Compost Fairie. In May, focus was on assembling TF 5 and 6. In June, work on TF 4, 5, and 6 continued with TF5 expected to be ready in 6 weeks, TF6 in November, and TF4 in spring. In June, removal of TF4 media was started.

In July, a startup schedule for all six TF was finally announced. Three contractor employees were injured on 7-29-204 causing broken legs when a plug came loose on TF3, probably related to build up of gas pressure during abrupt temperature change at sunrise. In July, start-up of TF5 and TF6 was initiated with TF1-4 in various stages of construction. In August, TF5 finally went on-line on 8-15-2024. In September, old media was removed from TF1. In September, TF 1-4 were in various stages of construction while TF 5 and 6 continued working. TF 4 should be functioning by 3-1-2025. In October, the removal of TF1 media was completed. TF 5 and 6 remained in service with work on the other 4 continuing. CROM applied sealant to TF1 effluent trough to replace the previously applied coating that sloughed off. BioAir units were installed. In November, it was announced that TF 4 should be ready by end of February 2025 with work on TF 1-3 continuing. By the middle of November, multiple cranes were working on TF 1,2,3, and 4 to expedite the CMAR project. In November, the removal of TF3 media was completed. The remaining covers and plates for TF4 troughs were installed. Shotcrete was applied to interior of TF4 walls. No decision yet made on TF mulching. TF4- the first fully rehabilitated trickling filter-expected to be ready for full service by end of February 2025. In December, the arms were put on TF 4 and pan testing scheduled. By the end of the year, it became fully obvious that there were not enough bio-pumps online to run the trickling filters. No TF recycling had been going since rehabilitation of the trickling filters began which is reason for lack of substantially improved performance. Recycle should start once 4 TF are on-line.

WIFIA Projects

The EPA announced the closing on the WIFIA loan on 10-8-2020 for $156 million which kicked off the start up of the new Admin Building, Lagoon 2A remediation, Primary Pump Station Upgrade, Secondary Dewatering Upgrade and Lagoon 5 expansion projects.

The year began getting new furniture for Admin building and working on punch list. In May 2024, Emerson continued trying to locate the missing 300 I/O points when the new Admin Building Data Center installed. In May, waiting on the Gate/Card reader continued. Work on the punch list items continued. By July, Emerson finally located missing 300 points for DCS to complete move of DCS to new Admin Building. On 7-29-2024, demolition of the original Admin Building began as Maxson Facility personnel had to adjust to less office space and less storage but more hallways and more open floor plans than accustomed to. On 8-5-2024, the 23 kV conduit was encountered buried beneath the floor of the demolished old Admin Building requiring special attention. On 8-26-2024, there was a meeting with Emerson regarding replacement of 300+ missing control points. In addition, the controllers at Blower Control Room would need to be upgraded to 64 bits to be compatible with the current version of Ovation. In September, paving of the parking lot was initiated and completed in October. The project was completed by November.

Construction of the pump building for the Lagoon 2A Phase 2 project continued from the previous year and new pumps were installed with the pumps at the crane load limit. In early June 2024, the lagoon cover was put back on. In July, CDM issued substantial completion for Lagoon 2A project enabling the Maxson Facility (in theory) to start wasting activated sludge into it as well as primary sludge. On 9-16-2024, CDM Constructors brought in Lord & Company to connect motorized valves on L2A and bring back to Ovation. CDM Constructors returned Lagoon 2A to the City on 10-2-2024 for sludge discharge operations with final completion expected on 11-22-2024. Work on punch list items continued during October. Damaged fiber

was repaired in November. Lord & Company worked with Emerson on programming. All pumps were working with Lagoon 2A recycling. Project was completed by end of 2024.

The Mayor executed the contract awarding the Pump Station Upgrade to Chris Hill Construction early in the year. A preconstruction meeting was held in March 2024 with mobilization beginning in May. The project was delayed because the MCC footprint was too large so the design had to be changed. The second half of the year was spent mobilizing, waiting for parts, and clarifying specifications.

Work on the Secondary Dewatering Upgrade continued from the previous year by Chris Hill Construction. Natural gas was extended to the facility for heating in January 2024. The belt filter presses were ordered in February, delivered in June, and then installed along with the rest of the equipment except of the conveyor system. The conveyor belt was finally installed in November 2024 after delayed delivery. Testing is expected to begin in January 2025 with startup after commissioning in March 2025.

Lagoon No. 5 Expansion design-related activities carried over from the previous year. Black & Veach performed a site visit on 2-6-2024. Contract documents were reviewed in February with plans to bid in May. In July, it was decided to wait until the Secondary Dewatering and Lagoon 2A Upgrade projects to be completed before bidding with both of those projects wrapping up by the end of the year. Efforts continued to be made to reach 100% design during the second half of the year. WIFIA approval for this stage of the project was received in about September. Bidding kept being delayed due to continuing changes to the design and specifications. As of December, the project had not been bid.

Biogas Quality Investigation

In 2024, Hydrogen Sulfide concentrations averaged 6253 ppmv based on 108 sample(s). Energy values averaged 536.8 BTU/cubic feet (dry) based on 108 sample(s). Average annual values of analytical results from previous years are summarized as follows:

Year	Soloxanes (ppbv)	Hydrogen Sulfide (ppmv)	BTUs/cubic foot (wet)
2013	124	19,820	468
2014	156	4591	441
2015	152	2836	483
2016	NA	935	320
2017	NA	1981	420
2018	NA	4540	350
2019	NA	1075	410
2020	NA	2012	362
2021	NA	2364	436
2022	NA	2433	253
2023	NA	105	388
2024	NA	6253	536.8

TVA Conversion to Natural Gas Related Projects

In August 2017, installation of the 5 new flares was completed by Kiewit Construction under TVA's contract and flaring of the waste gas using the new flares initiated. In September 2017, installation of the 4 bioreactors was completed and the reactors charged with media. Kiewit's contract to complete the job ended 11-5-2017 and operation turned over to the City. The City continued to work with CDM Smith and Biorem to work out warranty issues related to construction as well as Biorem's obligation to achieve steady-state operation so that maximum hydrogen sulfide levels do not exceed 480 ppm per the TVA contract with the City. On 12-12-2021, Biorem completed their performance trial. The system was able to meet contract specifications with untreated biogas H2S concentrations up to 10,000 ppm.

The revenue contract was signed with TVA on 1-3-2018 with a 20-year term effective once biogas treatment is adequate. The estimated capital contribution is $20 million based on a 6,500,000 MMBtu minimum agreement volume with TVA concurrently agreeing to pay the City 33 cents per MMBtu spread out over a 13-year payback period. This is based on a minimum annual average 500 BTU wet (higher heating value) and maximum monthly average Hydrogen Sulfide concentration of 30 grains per 100 CF or 480 ppm.

The City issued a Notice of Transfer of Biogas to TVA to begin 2-1-2022 and began sending production data to TVA semi-monthly. No biogas was produced during the year until June 2024. It peaked in July and dwindled down to zero by December. The SWG100 biogas analyzer, calibrated 7-28-2023, was used intermittently until the next calibration on 9-9-2024. The blowers were shut off on 8-5-2024 due to low biogas concentrations and high O_2 levels beneath lagoon cover #1. No biogas was flared during that period. The blowers were placed back in service on 8-12-2024. The blowers were shut off on 9-22-2024 due to low biogas concentrations and high O_2 levels beneath lagoon cover #1. No biogas was flared during that period. The blowers were placed back in service on 9-24-2024. Biogas was treated through the BioRem system and flared during 10-4– 10-5-2024. On 10-6-2024, the control system at BioRem experienced electrical issues and was shut down. No biogas was treated or flared from 10-6-2024 to 10-29-2024. Corroded flow meters were replaced and the PLC processor were also replaced in October 2024. Biogas was treated through the BioRem system and flared during 10-30-2024 – 10-31-2024 due to gas build-up under lagoon covers #1 and 2c. Biogas was treated through the BioRem system and flared on 11-1-2024 due to gas build-up under lagoon covers #1 and 2c. The BioRem was shut down again on 11-2-2024 due to low gas build-up under the lagoon covers. No biogas was produced, treated, or flared from 11-2-2024 – 12-31-2024. The water was drained from the system in December.

TVA continued to dewater pore water and stormwater from the east ash pond with treated stormwater discharged through outfall 002 which goes to the cooling channel. Startup of discharging treated pore water to the POTW began 3-10-2023 with 25.5 million gallons treated and discharged to the Maxson Facility to date. West ash pond restoration was completed on 11-7-2024. Total load out was 512,900 CY to the landfill. Ash was removed from the east ash pond and backfilled. Total load out through November 2024 was 747,600 CY to the landfill with completion expected in November 2029. The 36-inch bypass to MH23 was put in service on 9-14-2022. The permanent sewer line is expected to be completed in May/June 2025. After in service, temporary bypass line will be removed. Deconstruction/demolition of the Allen Fossil Facility began in about August 2022 with expected completion in 2025.

Biosolids Long-Term Planning

The Maxson Facility began operations in March 1975. Lagoon #1 was constructed along with the main plant. Lagoon #2 was then constructed as a supernatant clarifier in 1977-78 with supernatant returning to the head of the plant along with Lagoons #3, #4, and #5. TDHE approved the 160 acres south of Lagoon #1 for lime stabilization. The City sent a draft interim sludge handling plan to TDHE on 5-11-1978. The Consent Decree entered on 7-28-1978 had a section requiring the City to initiate an interim solution for "sludge handling" facilities. Pursuant to the Decree, the City built an 8-acre pond (Lagoon #3) and proposed to spray-irrigate the sludge on an adjoining field in the dry weather months. On 12-1-1978, the City began discharging sludge to Lagoon #3 which was anticipated to provide storage capacity for one year. As of 4-10-1979, the lagoon was at capacity and sludge began to flow back through the plant. The Second Amendment to the Consent Decree entered on 5-7-1979 allowed the City to intermittently discharge sludge through the plant's bypass until there was sufficient capacity to store sludge until startup of the land application program later that year. A revised sludge handling plan was submitted to TDHE on 7-2-1979. This plan stated that lime stabilized sludge would be applied to the 110 acres just south of the anaerobic lagoon during summer of 1979 with approximately 350 acres of additional land available adjacent this property for future use. On 1-2-1980, the City submitted a final revision of the plan that focused on sludge land application. By 5-6-1980, the City received approval for a sludge spray irrigation system on the 110 acres (west of the lagoons) with an additional 310 acres available for future use. The City acquired two sludge trucks with 4000-gallon tanks in May of 1980 and received a permit to apply sludge to a 310-acre dedicated site adjacent to the lagoons in June of 1980. By that point, a total of 280 million gallons of sludge storage capacity was available for stabilization and further digestion, followed by application on approximately 420 acres of dedicated land. On 4-21-1981, the Director of Public Works informed the City Council that the City had 608 days of sludge storage capacity left at the Maxson Facility and 375 days left at the Stiles Facility.

Sludge reports submitted to TDHE from 1980 to 1992 documented quantity of sludge applied and analytical results and as time went on, surface water run-off analytical results. Four sludge application sites were approved for the Maxson Facility beginning with Site 1 (110 acres) in 1979, followed by Site 2 (310 acres), in 1980, Site 3 (180 acres) in 1983, and Site 4 (3500+ acres) in 1982. Most of the sludge was applied to Sites 1, 2 and 3 at a rate of 15 to 20 dry tons per acre per year in an area that became known as the Earth Complex. Site 4 consisted of available agricultural land in Pidgeon Industrial Park where sludge would be applied at lower rates as fertilizer but apparently never extended beyond a nearby 500-acre site leased by a farmer. In September 1987, TDEC transferred all regulatory authority for municipal sludge disposal facilities from SWM to WPC in accordance with the Federal Water Quality Act of 1987 amendments. New NPDES permits effective 1-29-1993 incorporated annual and monthly reporting requirements for sludge management practices under Other Requirements.

The Maxson Facility added three primary clarifiers that became functional in 1982, resulting in the production of primary sludge. Initially, the primary sludge was sent to Lagoon #1. Sludge handling facilities for primary sludge were constructed and started up during the summer of 1983. These facilities included three belt filter presses with a total capacity of 100 dry tons per day and mixing with lime for stabilization. The dewatered primary sludge was then land applied

to Site 3 beginning about 1984. This practice continued until April 2004 when all primary sludge began being wasted to the lagoons for co-digestion with waste activated sludge.

During his last term in office, Mayor Dick Hackett on 8-23-1989 announced the creation of the EARTH Complex (Environmental and Resource Technology Complex) in the vicinity of the lagoons combining recycling with construction of a new landfill. An advisory board appointed by the Mayor would plan the development of the 4500 acres of land including a land fill (1250 acres) and a wildlife preserve (770 acres). One of the lagoons would be covered to recover methane. A wood mulching/composting facility was subsequently created as well as a sod farm and pumpkin patch visited annually by elementary school children. Long term plans included the construction of a management office and auditorium. While individual elements of the plan were eventually implemented either at the site or elsewhere, the overall concept was never implemented following the election of a new mayor in 1992, the promulgation of the EPA 503 regulations in 1994, and the focus on main plant expansion to accommodate industrial growth.

Digested sludge disposal continued through 1993 by spreading, injection, or spraying liquid 1-2% sludge onto the fields by tank truck. New federal regulations known as 40 CFR 503 became effective 2-19-1994 which included Surface Disposal of sludge that met appropriate vector attraction reduction and metals criteria. The City opted to comply with federal Surface Disposal Site requirements that included containment of surface run-off and ground water monitoring and so were no longer regulated by the state program. Berms and ditches were constructed to direct surface run-off and supernatant from the approximately 600 acres of dedicated land into the Nonconnah Interceptor via a gated 18-inch line leading into a manhole on the 84-inch Horn Lake Interceptor. Lagoon #1 was covered 1995 and the flaring of biogas began. Secondary Dewatering was placed in operation in 1996 based on an umbilical system patterned after the Stiles Facility except that the two lagoons providing feed sludge were not covered. Dewatered sludge was surface disposed in the new dedicated site. The use of the sludge spreading equipment was gradually phased out and replaced with dump trucks and dozers.

An anaerobic digestion study was done in 1998 to optimize current lagoon operations, including the direct wasting of primary sludge into the lagoons. Design of new covered Lagoons 2A, 2B, and 2C soon followed and constructed in 2001-2003. Plans were made to cover Lagoon #5 later. Sale of biogas to TVA began 2000 based on a 20-year agreement. The new covered lagoons were placed in service in spring of 2004 and the plan to waste primary sludge to the lagoons was enacted, ushering in a 12-year period of double biogas production. TVA did not properly maintain their equipment and were unable to utilize a great deal of the biogas the Maxson Facility produced. Lagoon #1 cover was replaced in 2011 due to cover failures dating back to 2007. Figure 5 shows long-term accumulation of Maxson Facility surfaced disposed biosolids and sludge accumulation within the lagoons. As shown, the new Secondary Dewatering practices implemented in the mid-1990s held down lagoon sludge accumulation until 2004 when primary sludge dewatering was suspended.

The sludge accumulation rate began increasing in 2004 when primary sludge dewatering was suspended and wasting of primary sludge to the covered lagoons for co-digestion with activated sludge began. Projects were implemented to increase sludge dewatering including the refurbishment of belt filter presses (2005), the purchase of three new additional belt filter presses (2013), the replacement of the two lagoon dredges with new units (2013), and the purchase of a third dredge (2019) to mine sludge from further out in Lagoon #5. The belt filter presses were

replaced in 2024 with commissioning expected in 2025. Hopefully, the current umbilical system has the capacity to keep up with the increased sludge production due to the co-digestion of primary sludge with waste activated sludge and an expansion of Secondary Dewatering will not be necessary.

Hydrogen Peroxide Application Pilot Testing in Lagoon No. 5

In 2020, the City contracted with USP Technologies (USP) to provide odor control at the Maxson Facility sludge lagoons using hydrogen peroxide (50%) dosing of digested sludge pumped to lagoon #5, which would reduce the emission of odorous sulfur compounds, especially hydrogen sulfide. USP designed and installed a hydrogen peroxide above-ground storage tank and dosing equipment (i.e., automatic dosing system). USP also provided on-going O&M and monitoring, hydrogen peroxide, periodic status reports and a web-based project dashboard.

The automatic dosing system uses a metering pump to continuously inject hydrogen peroxide into the digested sludge stream anytime the sludge pump is turned on. The metering pump is controlled by a sludge pump relay to turn both pumps on and off in concert.

USP periodically monitors dissolved hydrogen sulfide in the hydrogen peroxide treated digested sludge and gas-phase hydrogen sulfide at the digested sludge discharge point and a nearby fence-line to measure the performance of the odor control system. The hydrogen sulfide target levels at the control points are <5 mg/l of dissolved hydrogen sulfide in the treated digested sludge, <10 ppm of hydrogen sulfide vapor at the sludge discharge point, and <1 ppm at the nearby fence-line.

In calendar year 2024 a total of 21,170 gallons of 50% hydrogen peroxide was used for odor control at the Maxson Facility covered sludge lagoons at a total cost of $105.215. USP Systems have run without issue or incident. A few Acrulog issues occurred and resulted in some data gaps during the year, but were resolved asap. At the nearby fence-line the average hydrogen sulfide was 0.8 ppm and peaked at 19.6 ppm.

Interceptor Sampling Review

Tioga began collecting interceptor samples in November 2022 at weekly intervals to assess loading on each interceptor for comparison with influent loading. The sample locations also allowed for insights regarding pH and Metals. The Nonconnah Siphon downstream manhole (NO 020114) would include Blues City Brewing, Bioenergy, Kelloggs, Richardson Oil Seed, and Solae. This manhole would not include AB Mauri, Riviana, Valley, and Valero. These industries would show up in MH9. The results for 2024 are shown:

BOD ppd	Sampled Maxson Facility Loading	Nonconnah Interceptor MH9	Horn Lake Interceptor	Nonconnnah Interceptor NOI-S	Calculated Maxson Facility Loading
Jan-24	345,902	NA	10,171	304,393	382,531
Feb-24	571,774	NA	8,721	279,447	358,564
Mar-24	503,736	NA	9,589	267,747	338,852
Apr-24	642,414	NA	9,280	201,246	275,148
May-24	557,929	NA	11,266	224,792	296,774
Jun-24	476,356	NA	10,919	221,137	283,970
Jul-24	478,324	NA	11,431	225,438	289,552
Aug-24	343,274	NA	6,693	191,553	244,082
Sep-24	361,956	NA	NA	NA	NA
Oct-24	333,617	NA	NA	NA	NA
Nov-24	356,118	NA	NA	NA	NA
Dec-24	290,938	NA	NA	NA	NA
Average	438,528	NA	9,759	239,469	308,684
Percentage	100%	NA	2%	55%	70%

The calculated loading was based on a combination of Nonconnah loading (MH9 or downstream siphon), Horn Lake Interceptor, and industrial user loading (when not already included in interceptor loading). Compared to Sampled Influent BOD Loading, on average, Horn Lake was 2%, downstream siphon was 55%, and Calculated Influent Loading was 70%. No recycle/return stream impact was assumed. Sampled Influent BOD loading appears to overrepresent actual loading conditions in this data set.

TSS ppd	Sampled Maxson Facility Loading	Nonconnah Interceptor MH9	Horn Lake Interceptor	Nonconnnah Interceptor NOI-S	Calculated Maxson Facility Loading
Jan-24	218,083	NA	7,928	194,166	247,925
Feb-24	339,880	NA	7,754	134,279	193,586
Mar-24	427,008	NA	8,163	107,688	156,724
Apr-24	719,475	NA	9,842	96,299	144,898
May-24	460,326	NA	10,648	80,058	129,282
Jun-24	484,921	NA	15,561	104,806	149,425
Jul-24	669,377	NA	10,387	127,779	169,398
Aug-24	314,251	NA	10,090	48,372	84,758
Sep-24	399,319	NA	NA	NA	NA
Oct-24	204,205	NA	NA	NA	NA
Nov-24	265,629	NA	NA	NA	NA
Dec-24	233,937	NA	NA	NA	NA
average	394,701	NA	10,047	111,681	159,500
Percentage	100%	NA	3%	28%	40%

The calculated loading was based on a combination of Nonconnah loading (MH9 or downstream siphon), Horn Lake Interceptor, and industrial user loading (when not already included in interceptor loading). Compared to Sampled Influent TSS Loading, on average, Horn Lake was 3%, downstream siphon was 28%, and Calculated Influent Loading was 40%. No recycle/return stream impact was assumed. Sampled Influent TSS loading appears to be greatly over-reported in this data set.

External Metals Investigation

The Maxson Facility started seeing elevated levels of Molybdenum in 2022 in biosolids. As a result, an investigation was performed looking for potential external sources of Molybdenum as well as other pollutants. Samples were collected of influent, Horn Lake Interceptor, Presidents Island Interceptor, Nonconnah Interceptor at MH9, and Nonconnnah Interceptor downstream of the siphon near I240 at various times from 11-7-2022 through 8-12-2024. Sampling was a combination of grab and time-weighted composites. The purpose of collecting samples from the two different Nonconnah interceptor locations was to isolate industries coming in upstream of the siphon. Self-monitoring report and control authority sampling data were also reviewed and compared with interceptor and influent sampling results. The major industries reviewed are listed as follows in relation to the Nonconnah Interceptor sampling location:

Upstream of Nonconnah Siphon

- Blues City Brewing
- Bioenergy
- Richardson Oil Seed
- Kelloggs
- Solae

Downstream of Nonconnah Siphon

- A.B Mauri
- Riviana
- Valley
- Valero
- Valicore

Based on influent sampling, the metals of concern were determined to be chromium, copper, molybdenum, nickel, and zinc. A mass value at each sampling location was developed by multiplying average concentrations times typical flows. The results are as follows:

	Chromium ppd	Copper ppd	Molybdenum ppd	Nickel ppd	Zinc ppd
Nonconnah-downstream of siphon	1.2	19.4	1.44	10.56	46.7
Nonconnah-MH9	1.8	23.8	9.2	45	78
Presidents Island	0.2	1	0.01	0.24	3.1
Horn Lake Interceptor	0.01	1.2	0	0.01	3.84
Influent	8.74	60	14.8	77.5	197

The investigation was hindered by the fact that only one sample for metals was collected at Manhole #9 because the interceptor collapsed near that location and was under repair for most of the investigation. The industries discharging upstream of the Nonconnah Interceptor siphon are compared with the downstream siphon sampling location as follows:

	Chromium ppd	Copper ppd	Molybdenum ppd	Nickel ppd	Zinc ppd
Blues City Brewing	0.056	NA	NA	NA	0.544
Bioenergy	0.065	NA	NA	5.81	0.17
Richardson Oil Seed	0.049	0.24	NA	NA	0.68
Kelloggs	NA	0.086	NA	NA	0.212
Solae	NA	2.66	NA	1.44	9.96
Subtotal	0.17	2.986	NA	7.25	11.566
Nonconnah-downstream of siphon	1.2	19.4	1.44	10.56	46.7
Percent Explained by Industry	14%	15%	NA	69%	25%

The industries discharging downstream of the siphon and upstream of MH9 are shown below along with MH9 sampling results:

	Chromium ppd	Copper ppd	Molybdenum ppd	Nickel ppd	Zinc ppd
AB Mauri	0.027	0.15	0.167	0.102	1.689
Riviana	0.017	0.186	0.194	0.078	0.533
Valley	NA	NA	NA	NA	NA
Valero	0.065	0.327	1.244	0.388	5.19
Valicore	NA	NA	15.9	0.106	0.253
Subtotal	0.109	0.663	17.505	0.674	7.665
Nonconnah-MH9	1.8	23.8	9.2	45	78
Added between siphon and MH9	0.6	4.4	7.76	34.44	31.3
Percent Explained by Industry	18%	15%	226%	2%	24%

Apparently, the jackpot was hit for finding the source of Molybdenum which explained 226% of the result added between the siphon and MH9. Valircore was determined to be the major source followed by Valero. As noted, only one sample was collected from MH9 during the investigation.

The overall percentage of influent mass results explained by industries along with the percentage of influent mass explained by the interceptor sampling is provided below:

	Chromium ppd	Copper ppd	Molybdenum ppd	Nickel ppd	Zinc ppd
Percent Explained by Interceptors	23%	41%	62%	58%	41%
Percent Explained by Industry	3%	6%	118%	10%	10%
Percent Explained by HLI	0%	2%	0%	0%	2%

The investigation suffered by having limited MH9 data. If more data were available, it likely would have increased the percentages closer to 100% of influent explained by interceptor sampling since 90% of the influent flow goes through MH9.

Many of the industries sample semi-annually for metals along with annual sampling performed by the Control Authority. Based on this review, many of the industries are potentially underreporting and as far as molybdenum is concerned, not reporting at all. It is recommended that the industries mentioned in this report sample monthly for metals and that molybdenum be added to their self-monitoring. It is also recommended that Valicore be investigated as a major source of molybdenum as well as Valero. Site limits could be established for molybdenum for both Valicore and Valero in the future based on their discharge trends and impact on Maxson Facility biosolids. It is noted that Horn Lake Interceptor metals were detected at similar mass levels as the Mud Island Interceptor metals. Both interceptors have similar flowrates and would be represented of domestic background.

Disinfection Operation Overview

Continuous PAA dosing continued during 2024 at the Maxson Facility at a cost of $7,511,815 or $20,580 per day. This equates to 27.1 cents per thousand gallons treated at an average dose of 6.2 mg/l based on influent flow. The average pre-final effluent COD was 411 mg/l based on City lab analysis of 24-hour composite samples and 435 mg/l based on PXC instrumentation daily average.

All things being equal, the percentage of the Maxson Facility O&M expenditures spent on disinfection should be a good metric whether the operation is being properly monitored and costs contained. Over time, as experience is gained and the process optimized, the percentage theoretically should be decreasing. Conversely, if increasing disinfection costs are resulting in reduced allocation for non-disinfection expenditures, this could be indicated. Annual disinfection costs along with total expenditures since the first year of disinfection are presented below:

Year	Total Expenditures	Disinfection Costs	Percentage
2021	$20,331,743	$2,317,939*	11.4%
2022	$30,035,390	$8,781,819	29.2%
2023	$24,746,751	$6,691,536	27%
2024	$25,796,287	$7,511,815	29%

*only includes Aug-Dec 2021 (5 months)

COD Surcharge versus Disinfection Costs

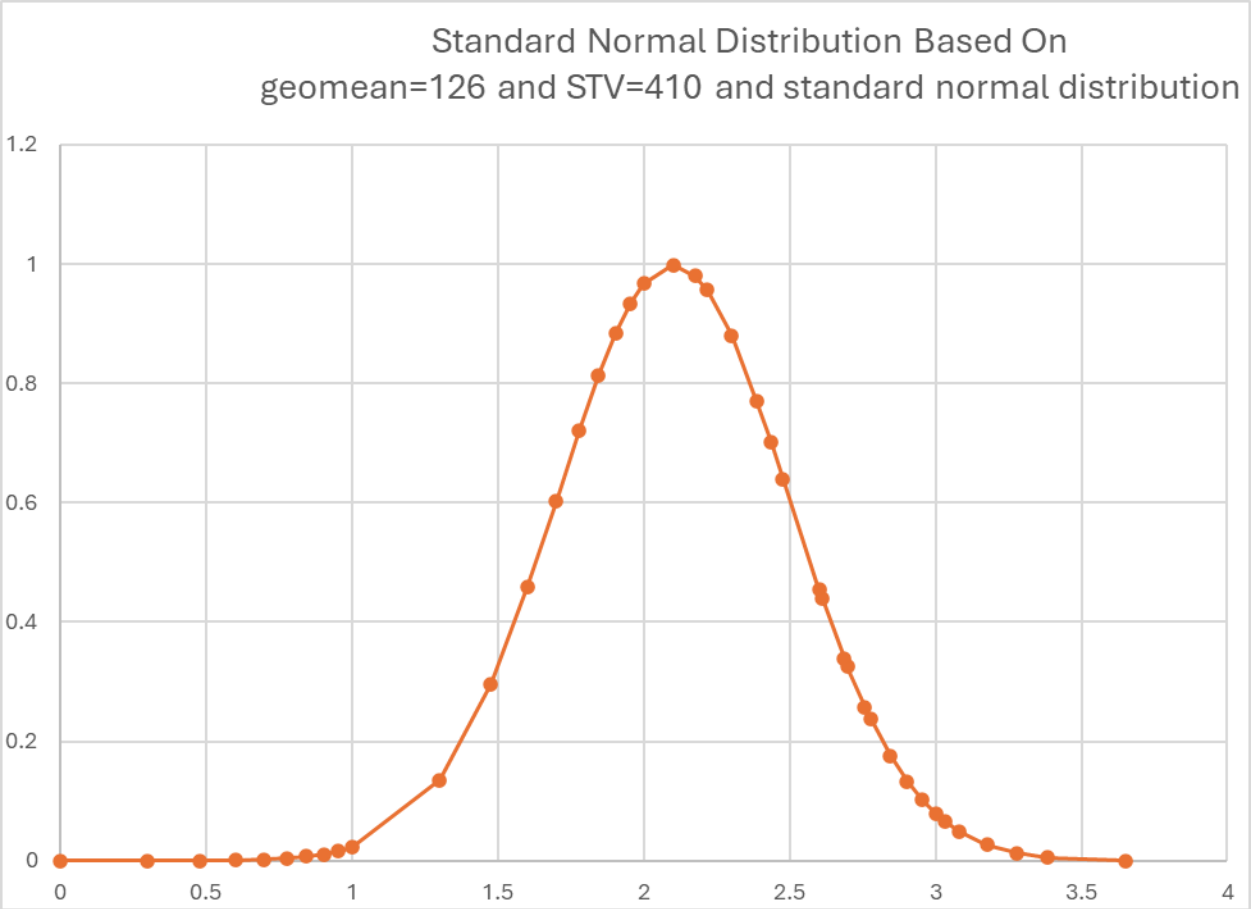
The 2024 COD surcharge for the industries that the Maxson Facility serves generated an estimated $7,312,435 in revenue. The 2024 disinfection cost was $7,511,815. The COD surcharge is expected to cover 50% of disinfection costs or $3,755,907. Only partial disinfection was performed. Assuming $10 million in disinfection costs for full disinfection, then 50% of disinfection cost would be $5 million. This would be a surplus of approximately $2,511,815 or 146%.

E Coli Statistical Review

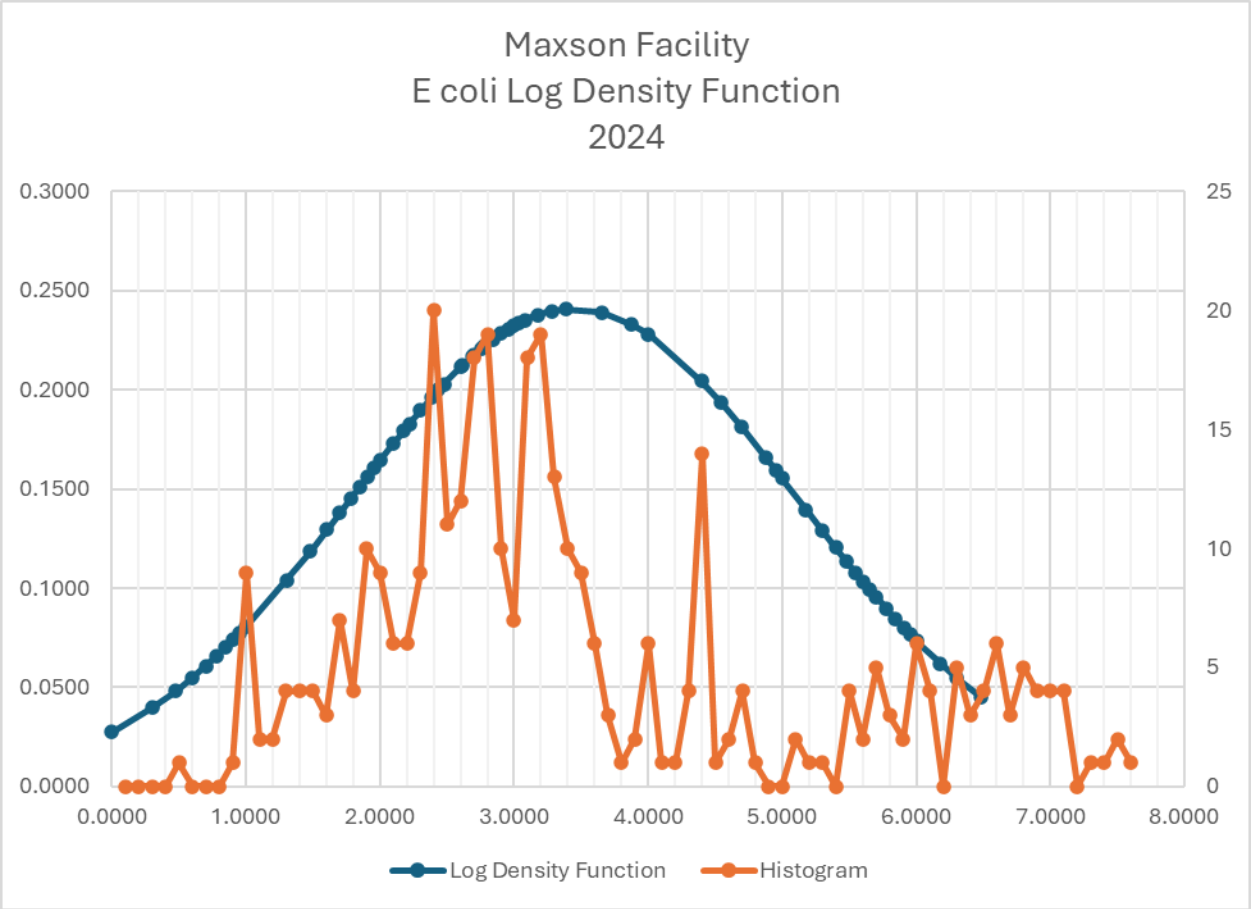
TDEC recently issued new regulatory requirements for disinfected E coli. For monthly reporting, the E coli geomean would remain 126 cfu/100 mls. However, the daily max would be 410 cfu based on 10% daily excursions. The previous daily max was 487 cfu. When questioned about the 10% daily excursions, TDEC claims that the 10% daily excursions is "built in" and not an allowance and that the 410 cfu be looked at simply as a daily max and not in conjunction with a statistical population.

EPA/FDA guidance, on the other hand, appears to suggest that the population curve for a 30-day period be calculated and the 90% statistical threshold value be calculated. This 90% threshold value is then compared with the 410.

The below graph shows the Standard Normal Distribution curve based on population geomean = 126 (log10 = 2.1) and standard deviation = 0.3996 based on Statistical Threshold Value = 90% or 410 (log10=2.612).



Below is the actual E coli data distribution for 2024. The actual data appears to be a mixture of two different data distributions centering around 630 and 1,600,000 E coli, respectively.



The following statistics are derived for the entire year:

Result	Percentile
126	8.5%
244	10.2%
410	11.5%
487	12%
62,887	50%

Based on the above table, there would be 321 daily max violations based on 487 daily max and 323 daily max violations based on 410 based on the hypothetical population curve. There were 221 actual daily max violations based on 487 daily max most of which were due to partial disinfection for PAA conservation during construction.

2.0 Stiles Wastewater Treatment Facility

Introduction-

It came to light that American Yeast stopped submitting monthly biogas usage in about March or April 2023 requiring further investigation. Stiles Facility began operating at 1 blower on 12-1-2023 initially due to only one fully functioning blower being available but soon stayed on one blower when another blower was available to reduce foaming coming out of the race ways and final channels apparently due to stressed biological conditions. These stressed biological conditions increased scum formation in the disinfection tank toward the end of December. Operators were unable to keep up with scum removal from the disinfection tank indicating it was atypical. To make matters worse, white, lard-like material also began entering the influent wet well and never able to be fully absorbed by the activated sludge [with PMC Biogenics and/or Rich Foods (via Onsite) being the prime suspects]. Memphis Cellulose had begun digestion about 10:21am on 1-4-2024. There appeared to be a an approximate 8-hour slug discharge associated with this start-up compounding problems associated with the process upset that began in December. This was followed by a series of daily maximum BOD concentration violations. Monthly Rich Foods Task Force meetings began on 1-10-2024 to gather data regarding this very atypical waste stream. Temperatures began plunging about this time due to sub-freezing highs accompanied by 6-inches of snow on 1-15-2024 MLK holiday. American Yeast stopped production on about 1-13-2024 after the initial hard freeze. On-Site shut down 1-12-24 through 1-23-2024. DuPont intermittently operated during 1-13-24 through 1-29-2024 and reported slug discharges 1-20-2024 and 1-21-2024. Runyon was down 1-12-24 through 1-22-2024. Memphis Cellulose was down 1-11-2024 through 1-23-2024. Pre-final color dropped 50% early 1-15-2024 as pipes began to freeze City-wide with a high temp of 12 degrees F. Pre-final effluent color was very choppy on 1-17-2024 and 1-18-2024 as American Yeast attempted to get back online. Pre-final effluent color began trending upward at 1:40 pm on 1-22-2024. There was an internal meeting on 1-23-2024 to review the 1-11-2024 slug discharge and daily maximum violation. The 1-11-2024 Stiles Facility sampling results showed influent COD = 1060 mg/l and influent BOD = 609 mg/l; and effluent COD = 368 mg/l and effluent BOD 142 mg/l. The effluent sample had strong ammonia smell indicating nitrification inhibition and likely other inhibition had occurred. As discussed later, increasing levels of Nickel and Zinc in influent and biosolids samples appeared to coincide with the symptoms of process upset (i.e. copious amount of foam) observed in December or perhaps before. Memphis Cellulose reported a slug discharge of 6000 gallons of magnesium sulfate on 1-24-2024. Pre-final disinfection was inundated with fibers on 1-26-2024 coinciding with the start-up of digestion at Memphis Cellulose. In addition, during the month, staff reported that B-side solids had been draining back to the head of the plant since the early 2000s possibly causing the solids imbalance between the A and B sides over the years favoring A-side with more solids. Comments to Brown & Caldwell's 11-30-2023 proposal to evaluate impact of industrial discharge on PAA disinfection and PAA disinfection optimization at the Stiles Facility were submitted.

Regarding the process upset, it was observed that zinc and nickel concentration in biosolids peaked in December 2023 after doubling since 2018. Effluent pre-final TSS and SVI were inversely related starting 11-1-2023 indicating the Stiles Facility was selecting out filaments (i.e. creating foam) due to stressed conditions. Stiles effluent BOD started getting erratic around 11-4-2023 with prefinal effluent TSS increasing after Thanksgiving as foaming increased and conditions continued to worsen. This resulted in more operational decline starting around 12-

12-2023 with increased foam production and floating scum in the secondary clarifiers and disinfection. IMPT reported elevated nickel in influent samples on 5-24-2023 and 5-25-2023 at 0.319 and 0.47 mg/l and elevated zinc at 0.462 and 0.314, respectively. Zinc levels began increasing in biosolids since July 2020 reaching a crescendo in December 2023. Nickel levels in biosolids began increasing January 2019. Zinc has an activated sludge inhibition threshold of 0.3 to 1.0 mg/l and a nitrification threshold of 0.08 to 0.5 mg/l. Nickel has an activated sludge inhibition threshold of 1 mg.l and a nitrification inhibition threshold of 0.25 mg/l. A similar situation occurred in 2010 related to PMC Biogenics which caused a process upset related to zinc. Nickel eliminates filamentous microbes and causes reduced SVI and sometimes used to control sludge bulking. In terms of Local Limits, there is a definite need to develop zinc and nickel inhibition limits based on actual conditions instead of relying on 2004 EPA guidance, especially for treatment plants without primary clarifiers.

At the beginning of February, the Stiles Facility was still recovering from the process upset that appeared well underway in the first half of January with indications that it started as early as November. The Stiles Facility began examining both influent and effluent COD to gain insight into these conditions. Memphis Cellulose was on-line 1-23-2024 through 2-3-2024 which corresponded to the on-going effluent BOD malaise. TDEC audited the lab on 2-5-2024. The second blower was turned on 2-5-2024 to address septic conditions after the foam had subsided. For the past 6 years, contact tank D.O. has peaked in January whereas in December 2023 and January 2024, it was reduced by 50% but still above 2.5 mg/l on average due to the atypical foaming. The Rich Foods Task Force met on 2-14-2024 in which it was learned that Onsite gets 30,000 to 50,000 gpd of Rich Products in heated form contained in 5-8 tanker truckers consisting mainly of Palm Oil which is very complex molecule and hard to break down. At one time, Denalli tried to land apply it but it would not break down. City and contractor crews continued to vactor a creamy material from clarifier raceways. Another effluent daily maximum BOD violation occurred on 2-8-2024. The Stiles Facility returned to one blower on 2-14-2024. It was learned on 2-23-2024 that the Stiles Facility EQ Basin project was not going to be pursued due to ground water interference. If an EQ Basin were to be built, the Dewitt Spain Airport would need to be relocated and the EQ Basin built above ground instead. During the month, CBS #2 fell to the bottom of the wet well, shearing the bolts, requiring the fabrication of a new bracket. It would remain out of service for the rest of the year due to hazardous atmosphere preventing access. BFP #5 rebuilt by Andritz was installed.

CDM Smith presented Moliere Nanobubble technology on 3-2-2024 focusing on reduction in organics to reduce PAA consumption. It is considered “clean chemistry” and does not replace FBD. On 3-18-2024, it was discovered that City IT was unable to keep the Stiles EDS in operation having lost all data since 12-8-2023. It was reported on 3-19-2024 that Power BI was unable to connect to recent Digi Data uploaded by the Smart Meters. John Abrerra had been working with Tioga for over a year and still not able to see the data on Power BI due to being stranded on an inaccessible Google Cloud account. On 3-22-2024, an EBS report was received for Stiles Facility Influent and RAS B samples collected on 3-5-2024. Concern was expressed with long chain fatty acids (LCFA) from palmitic and stearic acids and cations absorbing in the biomass. Zinc was detected at 3.36 mg/l in the RAS versus 0.34 mg/l in influent and Nickel at 0.22 mg/l in RAS versus 0.03 mg/l in the influent with ratios of 9.9 for zinc and 7.3 for nickel. LCFA sources include Rich Foods for Palmitic Products and PMC Biogenics for Stearic Acid. Stiles EDS started working again on 3-22-2024. On 3-28-2024 it was indicated that the Stiles Facility final channels were almost full of grit. This could be source of metals in recycle since

they are stored there was well as in the return sludge channel prior to secondary clarification. P1 Transformer at Grit Tanks/ Influent 480 V backup project was completed during the month.

On 4-1-2024, it was indicated that a 10:1 ratio of MLSS to influent for zinc and nickel was consistent in the literature indicating the metals to be easily absorbed to activated sludge. The EBS Report was reviewed on a 4-3-2024 with Katelyn Dreaux of BioWizard. Challenges at the Stiles Facility are palmitic and stearic acid, Nickle, and zinc. She recommended ORP measurements since D.O. is not reliable or accurate down to low levels. No significant surfactants or pesticide were observed in the test results. LCFA can be inhibitory if accumulating due to coating and inhibiting cell growth and likely to absorb on solids. Zinc was highest level seen in any client's RAS. She recommends soluble COD analysis across the entire system as well as ORP. As stated, LCFA and metals are the main issues. Palmitic and Stearic Acid were major contributors to elevated LCFA at the Stiles Facility. A treatability study could be performed on Nickel and Zinc. Zinc and Nickel likely caused formation of the billowing brown thick consisting of filamentous microbes and/or possibly polysaccharides. VFA would not necessarily show up in a FOG extraction; however, LCFA would. Data Master Plan kick-off meeting on 4-9-2024. Stantec began review procurement of a WIMS for future needs with a focus on disinfection tracking. Stantec also began work on comprehensive list of parameters to be tracked. High influent pH observed on 4-16-2024. Pre-final COD remaining elevated after process upset earlier in the year which has increased PAA demand relative to color. On 4-17-2024, it was speculated that the Stiles Facility needed to improve grit removal as ultimate solution to metals accumulation. Possible additional metals could have come from Front Street Interceptor cleaning (2019-2020), 120-inch interceptor cleaning (2021), and WR siphon cleaned (2023). As stated, pre-final COD relative to color appears to be creeping up while BOD going down. H2S monitoring with handheld devices began at the Stiles Facility wet well so that AECOM can complete their inspection. The wet well ventilation system crapped out several years ago so a 30% down payment was made on new ventilation system in February 2024. There was a meeting with CDM Smith regarding principal components analysis of 11-20-22 through 4-7-2024 disinfection-related data. PAA decay rate increases with temperature. Significant clusters of differences at the 24-26 deg C range were observed. PAA Total Dose, temperature, American Yeast COD and American Yeast TSS were all related by principal components analysis. Metals filings apparently caused very erratic readings of the influent 96-inch mag meter during the month. The fish and turtles were observed in MCS influent from mid to late April (starting 4-15-2024) when Mississippi River stage was above 21 feet. No foliage or limbs observed. The exact source of this inflow was not known during this time. Animals were intact indicating source is nearby.

Early in May, the Stiles Facility received the NOV/PAI from TDEC from the lab audit from 2-6 to 2-8-2024 with response due 5-24-2024. There was a meeting with CDM Smith on 5-13-2024 regarding nanobubbles technology. CDM will obtain their own nanobubble generator since Moliere unable to participate. The Mississippi River rose above 24-feet on 5-14-2024 with river stage = 21.8 feet being the level where flow increase was noticed. A slug discharge from American Yeast on 5-27-2024 caused reduction in scum (i.e. change in pH) and scum lines were kept open and working. On 5-28-2024, an NOV was received for the Stiles Facility January-February 2024 permit exceedances. The Stiles Facility received substantial completion for new Maintenance Building as the wait began for the mechanical contractor to complete punch list. Fine tuning on the new clarifier weirs continued and all troughs were removed and lowered by 7/8 inch and then sealed. H2S levels were still observed too high to inspect wet well for

condition assessment. Teems & Demovile was given the NTP for new wet well ventilation system.

At the quarterly biosolids meeting in June, it was learned that farmers were expressing concern over PFAS in biosolids from the Stiles Facility, being applied mainly to farmland in Shelby and Fayette Counties, used for hay and row crops. On 6-25-2024, it was observed that Chromium, Copper, Nickel and Zinc are consistently high in Stiles Facility RAS at 10X influent concentration. The contractor was still dealing with HVAC punch list items for new the Maintenance Building. Runyon's discharge seemed very low during June indicating a change in operations. PMC Biogenics removed their hydrogenation process on 6-21-3024 hopefully resulting in reduced nickel contribution to the Stiles Facility in the upcoming weeks.

On 7-1-2024, the Stiles Facility influent pH was elevated due to caustic leak at Memphis Cellulose. Preliminary review indicated PMC Biogenics to be a major source of both nickel and zinc and Runyon and American Yeast major sources of zinc. Calculations showed that 26,327 pounds of nickel currently stored in Stiles Lagoons 1,2, and 3 based on previous sampling data from 2019 or 1.8 pounds per day over 40 years. Similar calculations shows 640,016 pounds of zinc in Lagoon 1,2, and 3 or 43.6 pounds per day over 40 years. Both calculations give an indication of the amount of metals being received by the Stiles Facility on a daily basis. On 7-30-2024, there was a City-wide power failure that caused the Stiles Facility, including disinfection, to stop several hours. High pre-disinfection color readings occurred when power went back up as slug discharges inundated the Stiles Facility. Quarterly meeting with Material Matters was on 7-31-2024. Preliminary PFAS risk results for regulatory limits were discussed. Biosolids threshold appears to be 20 ppb on dry weight basis. PFAS in the 20-100 ppb range means to find the source and reduce application rate. City to receive PFAS questionnaire from EPA for TEM and MCS in a few months. Denali made plans to bring in a mobile press once new contract in place. There was a slug discharge attributable to American Yeast starting on 7-31-2024 and continuing all day 8-1-2024. Final adjustments to new weir on secondary clarifier A6 were made allowing it to be put into service soon. Contractor began demolition of the influent wet well old ventilation system. Slight inhibition evident in Stiles Facility process was evident since process upset at beginning of year with process taking longer to recover than expected.

On 8-12-2024, Jon Watson emailed photo of orange pre-final effluent with co-suspended cloudy solids, very atypical. The Base Dose continued to be increased relative to temperature due to increasing pass-thru COD. On 8-13-2024, the SCHD emailed regarding American Yeast request to increase ground water pumping with discharge of unused discharge to the "Stiles spray water system.' The ground water production permit provision was originally approved in 2001. SCHD appeared to be trying to get ahead of future questions from the public regarding demands on the aquifer. American Yeast ground water discharge follows distinct patterns. Since tracking began in November 2019, American Yeast discharge peaked in October 2021 at 113,211 gph before dropping to 46,854 gph in September 2023. It increased to 75,559 gph by end of 2024. American Yeast never confirmed this request with the Stiles Facility showing typical disconnect between industrial users and wastewater treatment facilities. On 8-26-2024, it was noticed that the Stiles Facility influent flowrate was pulsing every couple of hours. A detailed examination showed that influent flow began oscillating on 8-23-2024 at an amplitude of 15-20 at 1.5 to 2 hours cycle. This pattern was validated by observing Evonik's effluent flow. It was determined

that the backflow protection equipment for potable water needed to be replaced. Current potable is running from fire main cross over and not through the meter. The valve was shut off to water meter pending replacement. MLGW owns the back flow valve. As a result, the MLGW meter has been underreporting for several months. New BFP for Denalli arrived for installation. Arrangements were made to test American Yeast ground water for Iron and TDS to determine suitability as dewatering dilution and spray down. Investigation also showed that American Yeast biogas valve had been closed since 7-23-2023. Based on Tioga's observation, the clear orange liquid entering disinfection from time to time is originated from the Loosahatchie Interceptor.

Pulsations in Stiles Facility influent flow coincided exactly to the reported day that new VFDs installed at the Gayoso Lift Station on 8-23-2024. CDM Smith starting nanobubble testing on about 9-7-2024 concluding on 9-20-2024. They were able to collect data for a period when Memphis Cellulose was digesting (through 9-14-2024) and immediately after when they were not digesting. However, the data collected during the time after digestion was possibly confounded by the channel cleanout project. Changes were made to the programming of the Gayoso Lift Station VFDs on about 9-13-2025 to reduce the severity of oscillation but very little change in fluctuation at the Stiles Facility was evident. On 9-17-2024, Stiles Facility B-Side was shut down at 10am for final and return channel cleaning and bad diffuser replacement. This cleaning resulted in heavy color entering pre-final disinfection during periods of tank draining and heavy cleaning. B-Side shut down continued into October. A Pulsair webinar was presented by Bar Environmental on 9-24-2024. This mixing technology involves the release of extremely large air bubbles between 2 stacked plates to provide mixing across channel width which can be used to ensure proper mixing of PAA in the disinfection tank. The piping materials can be constructed of Schedule 80 PVC. Secondary Clarifier A6 new weirs were adjusted 3-4 times but still not correct. Red color liquid occasionally observed in Wolf River Interceptor during sampling performed was reported by Tioga during the month. Occasional orange color was observed in pre-final effluent was observed during the month when B Side was down (likely the same material).

The Stiles Facility returned B-Side to service on 10-3-2025. Ten (10) loads of activated sludge were brought from West Memphis (3000 gallons each) to seed the Stiles Facility biomass after the B-side channel clean-out to re-seed the biomass. Approximately 1,750,000 pounds (wet) of grit were removed during the channel cleanout, which included over 600 pounds (dry) of zinc and nearly 140 pounds (dry) of nickel. Denalli began operation of the mobile press on 10-7-2024. During the aftermath of B-Side cleanout, Stiles Facility RAS was reduced 75% since 9-17-2024 due to lack of build-up. It was learned that Evonik was caught off-guard in September due to side B shut down and channel cleanout because they originally expected lower PAA requirements with Memphis Cellulose not digesting. On 10-18-2024, it was observed that RAS B nickel and zinc concentrations were trending lower since last sample collected on 8-20-2024 before the cleanout. The grit tanks began foaming on 10-20-2024 with detergent type foam. On 10-21-2024, Stiles Facility staff began implementing 3-hour influent grabs in accordance with upcoming new permit requirements. On 10-24-2024, Evonik had bad control board causing occasional spikes in PAA. This was related to control of their gravity system and temporarily switched over to pumping system. During the month, the 1980 CH2MHill report was reviewed showing a maximum of 1.0 Contact f/m should produce a pre-final effluent sBOD of 8-14 mg/l. The current pre-final sBOD is 12 mg/l so things have not changed in 40+ years! However,

contact f/m has been trending upward and efforts should be made to return the contact tank f/m to its former <1.0 glory. The 1980 CH2MHill Report projected future wastewater loading and flow to be 135 mgd at 450,000 ppd or 400 mg/l. A full-scale test performed at the Stiles Facility indicated the future loading of 450,000 ppd could be accommodated operating 16 contact tanks and 12 stabilization tanks (including converting aerobic digestors) and running three blowers.

During the month, concern continued to be expressed regarding the low removal efficiency of grit tanks as evidenced by the tremendous quantity of solids stockpiled downstream in final and return channels downstream as well as elevated metal concentrations in RAS. On 11-15-2024, Hach was selected as WIMS provider based on screening during a selection period of several weeks. On 11-20-2024, Xylem was contacted regarding a valve management plan for 6-nch valves and larger. On 11-22-2024, the Side A of the disinfection tank was taken out of service revealing 18 inches of sludge accumulating between ½ and ¾ sampling points. B-Side which was also taken out of service for inspection showed no sludge accumulation. On 11-22-2024, the mechanical portion of the Stiles Facility wet well ventilation system was completed and the wait began for the next plant shutdown to complete the electrical as well as testing Circuit Scada Switches. It was reported that Runyon had been implementing new removal technologies in recent months which could explain the lower COD loading reported by them.

On 12-4-2024, the Stiles Facility composite sampling and interceptor sampling/flow metering kick off meeting was conducted. There will be a 4-feet deep vault between runways at DeWitt Spain Airport in place of the existing manhole over the Wolf River Interceptor. Stiles Facility effluent flow metering could possibly be improved employing the Flexim 721 which, according to the literature, can work on a 48-inch diameter RCP with 7-inch wall thickness. Flexim is owned by Rosemont which is a subsidiary of Emerson. On 12-6-2024, it was reported that Gayoso Lift station VFD computer programming was still being fine-tuned as influent flow rate continued to fluctuate at the Stiles Facility. On 12-7-2024, the pipe gallery flooded. A return sludge spool piece had been removed by a vendor working on A-2 on 12-6-2024. The B-2 effluent valve was leaking back from the Final channel (not fully closed). The 24-inch mud valve drain line was surcharged which backed into the tank. It backed into the inlet valve where the missing spool piece was. B-2 was also being drained which caused the 24-inch drain to surcharge. Influent was turned off 9 am to 5 pm to allow the final channel to empty. Lift Stations brought in two 6-inch pumps. MCS had 12-inch pump but no discharge line. Xylem brought in two 12-inch pumps mainly because they had sufficient discharge pipe. Once mud valve on B-2 was closed, they resumed using B-2 and sending sludge to the final channel. All transformers and a good deal of the 700 valve meters were lost. The Wemco pumps were returned to service on 12-13-2024 after repairs. A substantial amount of activated sludge was lost on B-side. The root cause of the incident was draining B-2 with B-2 effluent valve not fully closed. This caused sludge to start backing into A-2 which was being worked on which had previously been drained and mud valve open. The backup was due to surcharged 24-inch drain (mud valve) coming back into A-2. The 24-inch drainpipe was apparently surcharged. A spool piece has been removed from A-2 inlet valve through which the sludge back flowed. The contractor's forklift was also lost when the tunnels flooded out. An observation manhole was installed in 24-inch drain line during last plant pump around was possibly partially to blame. In the past, two tanks could be drained at once but not since the observation manhole was installed. A CCTV inspection would be required to confirm. Tioga observed that pre-disinfection temp went from 21 to 30 deg C when flow stopped but dosing continued possibly causing a reaction in the disinfection tank. During the month, it was observed that Stiles Facility influent flow (via

96-inch meter) was 8-10 mgd higher than expected. A new inverter was installed during April. The new Gayoso Lift Station VFDs were reprogrammed during the month in response to Stiles Facility requests. Flood/Lift originally reported that the erratic pumping was due to possible sewer overflow. Blower #1 bearing failed during the month. This blower may be down for a while due to difficult removal involved, requiring the construction of a gantry crane! Tristate installed the 2301 transformer at the blower building and began working on the installation of the second replacement. After several months of intermittent service, Stiles Facility was able to access a stable EDS at the main library via secure connection (i.e. wide area network). Planning startied on the second clarifier weir replacement. American Yeast groundwater discharge into the disinfection tank appeared to be causing the effluent temp probe to measure high requiring relocation of the probe. More attention needs to be paid to the American Yeast ground water discharge. Stiles Facility apparently removed booster pumps years ago from dewatering and other connections required for more beneficial use.

The Maintenance Department performed numerous noteworthy tasks during the year. In January, crews installed new lighting in the Air Main Tunnel to replace older lighting system. In February, a concrete pad and HVAC system for the Old Maintenance Shop was installed. In March, bollards were installed to protect Kerosene Tank next to Gas Tank behind Admin Building, #5 Dewatering Press was rebuilt, and a new water pressure gauge for Plant water was installed and connected to Ovation. A new 4-inch water line was installed to the Pump Building.

In April, the Drag-out chain and flights were replaced at the Grit Tanks. VFDs failed at Dewatering due to a power surge from lightning storm were replaced. In May, at Lagoon #4 (Surface Disposal Site), the outfall structure was demolished and replaced with manhole risers. In May, concrete on stairs leading down to Final Channel on A Side were repaired. A new 8-inch discharge Line for Barnes Pump in Pump Bldg was installed.

In August, at Disinfection, a new temperature RTD for Process use was installed and connected to Ovation. In September, at Side B Final Channel, all Down Pipes were removed and replaced with new ones. B7 Final Tank weirs and Scum Hopper were replaced. A new Gate Controller on the Main Gate was installed to replace failed older unit at the Admin Building.

In October, at the Pump Building, the Old Supply Make Up Air Handling Unit was demolished and replaced with a new unit. At the Blower Building, #1 Blower Motor was moved to #3 Blower Motor pad to get #3 up and going due to a failed motor. Mag Meters on Main Return Sludge line in West Tunnel were repaired. The concrete on stairs were repaired leading up to Grit Tank and Fine Bar Screens on the North and South Side. In November, at the Admin Building, a concrete pad was poured behind Lab around Chiller to protect from lawn mowing. In November, cracks in concrete walls in tunnel system were repaired with poly urethane as part of on-going repair. In December, all 0700 and 1600 mag meters in East Tunnel were repaired. The WEMCO pump motors in West Tunnel were replaced due to flooding. The Scum pump motors in West Tunnel were replaced due to flooding. Cracks in concrete walls in tunnel system were repaired with poly urethane as part of on-going repair. More spool pieces were installed in east Tunnel as part of on-going repair.

In addition to the major tasks listed above, the Maintenance Department completed 3597 Preventative Maintenance Work Orders, 389 Scheduled Maintenance Work Orders, and 3

Unscheduled Maintenance Work Orders for a total of 3989 work orders. The annual number of work orders completed over the last 10 years is summarized as follows:

Year	2024	2023	2022	2021	2020	2019	2018	2017	2016	2015
Preventative	3597	3063	963	NA	2853	3178	2795	3148	2694	2738
Scheduled	389	632	NA	NA	286	531	758	497	345	552
Unscheduled	3	11	354	NA	599	239	255	188	138	156
Total	3989	3706	1317	NA	3912	4013	3863	3857	3177	3446

Flow – The average daily influent flow for the Stiles Facility in 2024 was 86 mgd. This is a increase of 1 mgd from the previous year. The 0.5th percentile flow in 2024 was 67 mgd which is considered base flow for the year. The average daily flow from the major significant industrial users was 12.6 mgd. This leaves 6.4 mgd or 93% of average daily flow accounted for. The rainfall was 63 inches (plant gage). The average annual rainfall at the Stiles Plant gage has been 55 inches.

BOD_5 – The average influent BOD_5 in 2024 was 275,746 pounds per day. This is 12,787 pounds or 5% less loading from the previous year. The average influent BOD_5 concentration was 390 mg/l and the average effluent concentration was 43 mg/l for a BOD_5 removal efficiency of 89%. The average effluent BOD concentration adjusted for PAA was 34 mg/l for a BOD removal efficiency of 91%. The total average BOD_5 in pounds per day reported discharged by the major significant industrial users was 137,806 ppd in 2024 which equals to 50% of the total loading. This is 6589 pounds per day or 5% less from the previous year. The 0.5 percentile loading in 2024 was 108,682 ppd which is estimated domestic loading for this year. This leaves 29,258 ppd of BOD or 89% accounted for.

Total Suspended Solids (TSS) – The influent total suspended solids averaged 235,913 ppd in 2024. This is 4648 pounds per day or 2% less than the previous year. The average influent TSS concentration was 328 mg/l and the average effluent concentration was 24 mg/l for a TSS removal efficiency of 93%. The total average TSS reported discharged by the major significant industrial users was 58,245 ppd in 2024 which equals to 25% of the total loading. This is 1213 pounds per day or 2% less from the previous year. TSS loading from lagoon supernatant and dewatering filtrate return was sampled at 25,726 ppd or 11% of total average daily TSS loading. The 0.5 percentile loading for 2024 was 124,610 ppd which is estimated domestic loading for this year. This leaves 27,332 ppd of TSS or 88% of average daily loading ccounted for.

Chemical Oxygen Demand (COD) - The influent averaged 919 mg/l and 651,038 ppd which is 18,154 ppd or 3% increase from the previous year. The effluent averaged 149 mg/l and 106,869 ppd which would include PAA. The 0.5 percentile loading for 2024 was 221,017 ppd which is the estimated domestic loading for the year. The total average COD discharged by the major significant industrial users was 295,482 ppd which equals 45% of the total loading. This leaves the remaining 134,539 ppd or 80% of influent COD loading ccounted for. The COD surcharge is based on covering 50% of disinfection costs.

Effluent E. Coli - Effluent E. analysis was performed on daily effluent grab samples throughout the year as required by the NPDES permit as report only. Effluent E. coli results for this past year are presented in Figure 6 based on disinfected effluent.

Water – In 2024 the metered water usage totaled 106,405 CCF at a cost of $149,316. The average metered water consumption was 10,641 CCF per month or 0.22 MGD. This is a decrease of 3336 CCF per month from the previous year which indicates the flows are potentially being bypassed around the meter. Flows discharged directly from American Yeast for use by the Stiles Facility (i.e. theoretically for aeration basin and scum hopper spray) averaged 1.6 mgd. Ideally, this American Yeast ground water would be used for motor cooling and dewatering spray down.

Energy – The plant main power consumption was an average of 5,598,639 kwh of electricity per month in 2024. This is approximately 368,004 kwh per month more than the amount used in the previous year. The monthly peak demand averaged 8375 kw which is 221 kw more than the previous year. The average monthly main power bill in 2024 was $456,554. This is $37,098 more per month than the previous year. Payments for participating in the ENERNOC curtailment program for main power are not available. It is believed that the co-gen fired by natural gas and serve this function in the future.

November was the high month for demand at 8948 kw and October was the high month for consumption at 6,477,264 kwh. January was the low month for demand at 6253 kw and January was the low month for consumption at 4,139,616 kwh. Hourly electric demand throughout 2024 is shown in Figure 2.

Effective 10-1-2015, the City entered into an agreement with MLGW to switch from Seasonal Demand to Time of Use billing. TVA would charge Fuel Cost Adjustment based on user class. TOU rate structure benefits high load factor customers through incentivizing off-peak energy usage. The contract term is 5 years and will be automatically extended unless otherwise terminated. The City signed an alternate circuit agreement with MLGW on 10-2-2019 for 10 years for the circuit switches to be on the MLGW power poles. This will result in an additional $13,778.23 monthly charge on the main power bill once the program is in place.

Sludge and Biosolids – The Stiles Plant wasted 104 million dry pounds of secondary sludge to the lagoons in 2024. This is 6.1 million pounds less compared to the previous year. 19.8 million dry pounds was digested and dewatered at a 77% capture rate.

The annual amount of digested sludge (in million dry pounds) removed from the two covered lagoons over the last 10 years is provided below:

2024	2023	2022	2021	2020	2019	2018	2017	2016	2015
19.8	25.8	40.1	38.3	36.1	38.8	43.9	31	44	30

Approximately 36% of the secondary sludge wasted to the lagoons in 2024 was destroyed. The formula used to calculate percent destroyed is %VS of WAS multiplied by %VAR. This is 37.2 million dry pounds (18,600 tons). 8.9 million dry pounds was removed from Lagoon 3N by the contractor and disposed at the landfill. Zero million dry pounds was removed from Lagoon 3S

by another contractor and disposed at the landfill. Considering removal and volatilization the new change in the lagoon solids inventory was an increase of 29 million dry pounds within Lagoon Nos. 1, 2, and 3. The calculated cumulative dry solids within the three lagoons is 866 million dry pounds. In 2024, a net approximately 28 million dry pounds (14,000 dry tons) of bio-solids were removed from Surface Disposal Site and about 23.6 million dry pounds (11,800 dry tons) were surface disposed at the Maxson Facility, including newly dewatered sludge and surface disposed material combined. The calculated dry bio-solids that has been placed in the Surface Disposal Site since 1984 is 1357 million dry pounds. This is the second recorded net decrease in the Surface Disposal Site amount.

For the last 10 years, net accumulation has been occurring at an average rate 30 million dry pounds per year in Lagoons 1, 2, and 3. Sludge removal from the three lagoons has averaged 34 million dry pounds per year. To stop net accumulation, the Stiles Facility will need to dewater and surface dispose or land apply and/or clean out lagoons and dispose an average of 64 million dry pounds per year which is 213% the existing rate of removal or destruction.

The total cost for the dewatering and surface disposal/land application/landfill disposal operation was approximately $7,017,104. This equates to a unit cost of $155 per dry ton of bio-solids surface disposed, landfilled, land applied and volatilized in 2024. The annual cost per dry ton for disposal/land application and volatilization combined over the past 10 years is as follows:

2024	2023	2022	2021	2020	2019	2018	2017	2016	2015
$155	$208	$232	$142	$94	$104	$83	$93	$85	$102

Generally, the unit cost to dewater and dispose sludge as bio-solids is a function of polymer cost and dewatering throughput. The cost to dewater and surface dispose, landfill and land apply digested sludge was $262 per dry ton in 2024. The annual cost per dry ton based on dewatering throughput over the past 10 years is as follows:

2024	2023	2022	2021	2020	2019	2018	2017	2016	2015
$262	$315	$367	$233	$207	$186	$169	$250	$168	$129

Biogas- Approximately 567 million cubic feet of biogas was produced 2024. The annual biogas production over the last 10 years is as follows (in millions of cubic feet per year):

2024	2023	2022	2021	2020	2019*	2018*	2017*	2016*	2015*
567	398.1	391	396.7	349	172	167	148	151	275

Years marked * have been revised upward and are based on blower electrical consumption rather than flow meter readings due to faulty flow meter. The annual daily average bio-gas production over the last 10 years is as follows (in million cubic feet per day) based on revised data:

2024	2023	2022	2021	2020	2019	2018	2017	2016	2015
1.55	1.09	1.07	1.09	0.95	0.47	0.46	0.41	0.41	0.75

The biogas produced equates to 15.2 cubic feet of biogas per pound of volatiles destroyed. The yield (ratio of biogas to pounds of volatiles destroyed) during the 10-year period based on revised 2013-2019 data is shown below:

2024	2023	2022	2021	2020	2019	2018	2017	2016	2015
15.2	10.8	12.9	10.2	8	6.3	4.2	3.5	3.9	8.1

The amount of sludge volatilized each year appears to be constant. This could be the result of a fixed quantity of inert solids being dredged and dewatered and not associated with fresh input and biasing the amount of volatile reduction on the high side in these lagoons. It could be that a portion of the sludge is not actually being destroyed but is accumulating as an isolated volatile fraction that is remaining in the lagoon and not being dewatered. This isolated volatile fraction is not being properly mixed and broken up to allow anaerobic microbes to digest it due to short-circuiting and contributes to the reduction in viable lagoon volume. Trends in biogas production over the years are shown in Figure 7. The increase in biogas production beginning in 2021 is believed to be due to long periods of time between switching lagoons for wasting which allows for the breakup of the isolated volatile portion and increases biogas yield. Even though the quantity of amount of volatile reduction appears to be constant from year to year, the quality of the volatiles being produced (i.e. that can convert into biogas) has improved as the biogas yield has been constantly improving since 2017. This could be related to the practice beginning in late 2020 of dredging from the lagoon year-round as opposed to April-November only which was previous practice.

In 2024, American Yeast received zero million cubic feet of biogas from the Stiles Facility which is zero % of total biogas production. The revenue received from American Yeast for this biogas was zero $. The annual usage by American Yeast for the last 10 years along with the amount of biogas flared as waste gas by the Stiles Facility including revised 2014-2019 data is as follows (in millions of cubic feet):

Year	2024	2023	2022	2021	2020	2019	2018	2017	2016	2015
American Yeast	0	0	91.86	128.15	152	48.9	71.9	85.3	63.1	80.5
Flared	567	398.1	391	269	197	133	95.1	62.7	87.9	194.5

Permit Violations- This year the Stiles Facility had 42 NPDES Permit limit excursions related to treatment efficiency and performance.

In January, there 9 permit excursions. There were BOD monthly average concentration, monthly average percent removal, 1 weekly average concentration, and 6 daily maximum concentration excursions. The plant experienced a process upset in January as the plant recovered.

In February, there were 16 permit excursions. There were BOD monthly average concentration, 2 weekly average concentration, and 12 daily maximum concentration excursions. Upset conditions carried over from previous month. Influent and mixed-liquor samples showed elevated levels of nickel and zinc.

In September, there were 6 permit excursions. There were BOD monthly average concentration, 1 weekly average concentration, and 4 daily maximum concentration excursions. The excursions occurred when Side B was taken out of service to cleanout accumulated grit from the final and return channels

In October, there were 2 permit excursions. There were 1 BOD weekly average concentration and 1 daily maximum concentration excursions. The excursions were a carry over from the previous month as Side B returned to service.

In December, there were 9 permit excursions. There were BOD monthly average concentration, two weekly average concentration, and 5 daily maximum concentration violations. Tunnels flooded on 12-7-2024 related to work being performed on two aeration basins.

The Stiles Facility has had 1045 permit violations for treatment efficiency and performance since the state permits were issued in 1982. The cumulative violations are shown in Figure 4 with Maxson Facility cumulative violations for comparison. The excursions associated with different major periods of plant history are as summarized below:

Period	Average Number of Excursions Per Year	Description
1982-1984	106.7	Commissioner's Order
1985-1992	6.5	Municipal Compliance Plan Annual Reporting
1993-2010	14.5	Occasional collection system failures and high river stages
2011-2024	29.4	Increasing collection system failure, high river stages, and process upsets

The Stiles Facility has had 5 years (1987, 1992, 1999, 2001, and 2007) with zero violations. Excluding years of extended high river stage and/or major collection system failure, the Stiles Facility has averaged 2.2 violations per year (24 out of 42 years). The highest number on record was 2019 with 181 violations.

The pounds-based parameters were eliminated from the Stiles Facility permit beginning in 2011 with only concentration and percent removal parameters remaining for BOD and TSS. The Stiles Facility number of violations could have been higher had the pounds permit parameter not been eliminated.

Bio-solids Long Term Planning

In 1971, planning was initiated by the City Division of Engineering to turn municipal garbage into liquid fuel by processing it at a facility to be located at Airways and Democrat and piping the fuel to the TVA Allen Steam Plant. TVA subsequently turned down the proposal. In 1973, when construction of the Maxson Facility started, planning was initiated to incinerate both garbage and municipal treatment plant sludge and sell the steam to TVA. In 1974, the City executed a contract with Wegman Engineering to perform a feasibility study of the construction

of incinerators for this purpose. In the same year, MLGW announced plans that it would construct a garbage incinerator to provide steam for downtown buildings. In response, the City expanded its scope of incineration in April 1975 to include flammable liquids. The two competing proposals were submitted to the City Council for review and the MLGW plan was selected on 9-9-1975. In January 1976, in an apparent compromise, the City Council agreed to give half of the garbage to MLGW to generate steam for downtown office buildings and half to the City to generate steam for the Allen Steam Plant. The City was able to qualify construction of its proposed incinerators under 201 Facilities Planning and included it in the 1977 Construction Grants application submitted to the US EPA. Municipal garbage would be ground up and then trucked to the Maxson and Stiles Facilities for co-incineration with the sludge. MLGW abandoned its incinerator project in early 1978 when its consultant determined that the City had overestimated its garbage production. On 8-9-1978, EPA gave preliminary approval for the incineration project and City Council approved on 8-22-1978. However, on 10-6-1978, the EPA announced that it would only fund 33% of the cost or $49 million since that was the portion of the project related to municipal sludge and the City would have to pay the remainder or $130 million. The City had hoped the EPA would fund 75% as with other wastewater projects. In the meanwhile, sludge continued to accumulate in lagoons at both treatment plants, including Lagoons 1 and 2 at the Stiles Facility. The EPA gave the City a final deadline of 9-1-1979 to make a decision whether or not to accept the grant, indicating that potential enforcement would be taken if proper sludge disposal arrangements were not made. The deadline was passed without a decision being made.

The Stiles Facility generated 245 million dry pounds of sludge after the treatment plant first opened from 1978 to 1982. In 1980, the City designated a 53-acre area adjoining the south end of the airport runway as a temporary bio-solids landfill known in the early planning documents as Sludge Landfill No. 1. A scope of operation plan dated 4-14-1981 was prepared for this facility and approved by the TDHE. On 4-21-1981, the Director of Public Works informed the City Council that the City had 608 days of sludge storage capacity left at the Maxson Facility and 375 days left at the Stiles Facility.

During 1980 to 1983, the plant began dipping sludge out of the lagoons and disposing it in Sludge Landfill No. 1 of which only 35 acres was used. There was no dewatering performed at that time as the original dewatering facility was improperly designed so a second contract was issued to build a another. Lagoons No. 3 and No.4 comprising 65 acres were permitted by TDHE as a sewage sludge landfill on 1-18-1984 with requirements for groundwater monitoring and runoff control. This was known as Sludge Landfill No. 2. The City constructed Lagoon No. 4 and began using it in the summer of 1984. New federal regulations known as 40 CFR 503 became effective 2-19-1994 which included Surface Disposal of sludge that met appropriate vector attraction reduction and metals criteria. The City opted to comply with federal Surface Disposal Site requirements that included containment of surface run-off and ground water monitoring which the City was currently doing and so was no longer regulated by the state program. This became established as the Surface Disposal Site for the next 25+ years.

Under Pathogen Reduction for Surface Disposal, the plant has been reporting under Class B, Process to Further Reduce Pathogens, Alternative 2 (anaerobic digestion). Under Vector Attraction Reduction, the plant has reported under Alternatives 1 and 2 (greater than 38% volatile reduction, less than 17% additional volatile reduction) based on waste activated sludge

as the starting point and feed solids as the endpoint for Alternative 1. Prior to 2002, dewatered cake was the endpoint for Alternative 2. Beginning in 2002, feed solids were the endpoint for Alternative 2.

Figure 8 shows the calculated cumulative net change in material within Lagoon Nos. 1, 2, and 3 and the cumulative dry solids stored in the Surface Disposal Site. The Surface Disposal Site is at approximate full capacity and new biosolids are being diverted to land application sites and the Maxson Facility Surface Disposal Site.

Based on record keeping and volatile reduction calculations, the amount of material stored in the three lagoons is 362,000 dry tons. An additional 18-20% reduction of the original material may have occurred due to longer term volatilization given the extreme 35+ year holding period. This would put the in-situ quantity at 289,600 dry tons. Lagoon 3S is believed to be near sludge storage capacity based on an investigation completed during the months of November and early December 2016 with an average total solids content of 15%. Lagoon Nos. 1 and 2 can remain useable if the City can reduce the net accumulation rate. The City continued drying out Lagoon 3N using heavy equipment to gain use of the area as a future surface disposal site.

A proposed biosolids master plan was presented to the Director on 3-22-2023 focusing on options for the Stiles Facility Surface Disposal Site and using the existing 870-acre Maxson Facility surface disposal site be used for both Maxson and Stiles Facility biosolids. He emphasized the City's need to utilize the adjoining DeWitt Spain Airport property for Stiles Facility biosolids management as well as PFAS impact on land application. The additional land would certainly come in handy for solar drying material generated from lagoon clean-out prior to off-site disposal, etc

On 6-3-2020, a contract with Denali was executed to provide land application of biosolids on farmland and/or landfill disposal through 8-1-2023 with two additional 1-year extensions. Land application of Stiles Facility biosolids continued through 2024. During 2024, approximately 9.7 million dry pounds of fresh dewatered biosolids was combined with an equal amount of stored Lagoon 4/Surface Disposal Site biosolids and transported to the Maxson Facility for surface disposal. This practice was performed monthly with the exception of May and November.

Disinfection Operation Overview

Continuous PAA dosing began on 11-10-2018 at the Stiles Facility and continued through the year at cost of $16,894,680 or $46,287 per day. This equates to 54 cents per thousand gallons treated at a calculated average dose of 12.6 mg/l (based on influent flow). The average pre-final effluent color was 679 Pt Co based on City lab analysis of 24-hour composite samples and 792 Pt Co based on PXC instrumentation daily average (AIC 412).

All things being equal, the percentage of the Stiles Facility O&M expenditures spent on disinfection should be a good metric whether the operation is being properly monitored and costs contained. Over time, as experienced is gained and the process optimized, the percentage theoretically should be decreasing. Conversely, if increasing disinfection costs are resulting in reduced allocation for non-disinfection expenditures, this could be indicated. Annual disinfection costs along with total expenditures since the first year of disinfection are presented below:

Year	Total Expenditures	Disinfection Costs	Percentage
2019	$35,471,331*	$14,915,064**	42.1%
2020	$26,286,152	$10,626,522	40.6%
2021	$30,915,816	$12,680,787	41.0%
2022	$33,762,346	$10,975,110	32.5%
2023	$40,992,619	$15,048,244	36.7%
2024	$39,967,098	$16,894,680	42.3%

*includes Nov-Dec 2018 total expenditures
**includes Nov-Dec 2018 disinfection costs

Impacts on Final Effluent BOD

Continuous PAA dosing began on 11-10-2018. The actual PAA usage for 2024 and calculated dosage of active ingredient based on influent flow is provided as follows:

Month	Amount of 15.5% PAA Solution Used (pounds)	Calculated Average Dose of Active Ingredient (mg/l)	Increase in Soluble BOD Between Pre-Final and Final Effluent (mg/l)	Final Effluent BOD (mg/l)	Adjusted Final Effluent BOD (mg/l)
January	2,061,160	11.6	12	58	41
February	1,798,683	11.2	13	72	56
March	1,723,005	11.3	10	35	25
April	1,524,244	9.5	6	28	22
May	1,699,513	9.4	7	32	25
June	1,946,105	12.6	4	32	28
July	2,258,313	15.2	9	43	34
August	2,629,783	18.2	5	39	32
September	2,615,200	17.1	10	53	42
October	1,919,001	13.1	7	41	33
November	1,651,058	10.5	4	28	23
December	1,951,645	11.3	6	57	48
Average	1,981,476	13	8	43	34

Color Contributors and Impact on Disinfection

The Color Contributors impacting the Stiles Facility consist of Memphis Cellulose, Runyon, and American Yeast as explained in this section. The major differentiator in PAA dosage is whether Memphis Cellulose is online. The following statistics were derived for various parameters during 2024:

Parameter	Memphis Cellulose Down	Memphis Cellulose Running	p-value**
PAA Solution Usage ppd*	62,757	74,771	0.032
PAA mg/l*	17.9	20.3	0.045
City PAA residual mg/l	1.01	0.92	0.345

Influent COD mg/l	856	991	0.000
Pre-final Effluent COD mg/l	144	177	0.014
Final Effluent COD mg/l	137	165	0.189
Influent Flow mgd	84	89	0.330
Effluent Flow mgd*	72	64	0.072
E coli (99 percentile)	1,235	1,658	
E coli (geomean)	46	44	
Number of Daily E coli Permit Exceedances	7	5	
Influent Color-City	2668	2332	0.030
Influent True Color-City	463	901	0.000
Pre-final Color-City	587	781	0.000
Pre-final True Color-City	219	398	0.000
Pre-final Color (AIC 412)*	683	919	0.022
Pre-final Effluent sBOD mg/l	20	25	0.000
Final Effluent BOD	40	46	0.025
Increase in sBOD mg/l after Disinfection	7.4	8.2	0.000
Pre-final Effluent BOD mg/l	37	45	0.000
Days Down or Running	197	169	

*Based on Evonik data from Ovation
**P-values below 0.10 indicate significant statistical difference

Based on the above data, the impacts of Memphis Cellulose on Pre-disinfection effluent quality as pass through can be determined as shown on the following table for 2024:

	Influent Concentration Increase with Memphis Cellulose Digesting mg/l	Prefinal Effluent Increase with Memphis Cellulose digesting mg/l	Percent Pass Through to Disinfection for Memphis Cellulose
COD	135	33	24%
Apparent Color	-336	194	Unknown
True Color	438	179	41%

Influent Apparent Color showed a decrease with Memphis Cellulose digesting (2332 compared to 2668) whereas Influent True Color showed a definite increase (901 compared to 463). 24% of Influent COD contributed by Memphis Cellulose and 41% of Influent True Color passed through to disinfection.

The impact of all other industries, including American Yeast and Runyon, on Pre-disinfection effluent quality as pass through can be determined as shown on the following table for 2024:

	Discharge for Users other than Memphis Cellulose mg/l	Prefinal Effluent Discharge for Users other than Memphis Cellulose mg/l	Percent Pass Through to Disinfection for all other users
COD	856	144	17%
Apparent Color	2668	587	22%
True Color	463	219	47%

17% of Influent COD contributed by all other users, including American Yeast and Runyon and 47% of Influent True Color passed through to disinfection.

The impact of Memphis Cellulose on Final Effluent quality is shown on the following table for 2024:

	Memphis Cellulose Status		
Parameter	Not Digesting	Digesting	P-value
Final Eff BOD mg/l (adjusted for PAA impact)	33	35	0.083
Final Eff BOD mg/l (not adjusted for PAA imact)	40	46	0.025
Stiles Final Eff TSS mg/l	23	25	0.061
Days	197	169	

The impact of Runyon Industries on PAA dosage and impact of pre-disinfection effluent quality is readily becoming apparent. Runyon Industries is typically a day shift only, Monday through Friday operation. They are off on holidays but work some Saturdays and very occasionally work Sundays. To discern its impact, a data evaluation was performed with and without Memphis Cellulose on-line. Note that the weekend and holiday data which is when Runyon is typically off-line may be biased lower since other industrial users may have a similar schedule. The below table shows the impact of Runyon operations on influent and pre-disinfection quality based on Memphis Cellulose digestion status.

	Memphis Cellulose Not Digesting		Memphis Cellulose Digesting	
	Runyon Off Line*	Runyon On Line	Runyon Off line*	Runyon On Line
PAA Solution ppd	56,806	63,869	74,109	79,512
PAA dose mg/l	15.8	18.0	21.9	21.4
Stiles Final Eff PAA mg/l	1.2	0.9	0.6	1.0
Influent COD mg/l	713	930	849	1034
Influent Apparent Color	2498	2755	2168	2381
Influent True Color	378	507	941	890
Pre-final COD mg/l	131	151	173	178
Final Eff COD mg/l	112	149	158	166
Stiles Influent Flow mgd	82.7	84.2	86.1	89.7
E coli (99 percentile)	2134	812	2420	1271
E coli (geomean)	52	43	45	44
Number of E coli Daily Permit Exceedances	2	4	2	3
Stiles Pre-final Eff Color	554	604	906	744
Pre-final AIC 412 Eff Color	639	705	928	916
Pre-final True Color	193	233	426	390
Pre-final sBOD mg/l	14	23	20	27
Pre-final Eff BOD mg/l	29	42	38	47
Final Eff BOD mg/l	29	46	38	49
Increase in sBOD after disinfection	4.9	8.7	7.5	8.5
Final Eff Adj BOD mg/l	23	37	30	37
Days Running or Down	67	130	39	130

*Note that data with Runyon Industries off-line will be more representative of weekends and holidays.

Runyon appears to have an outsized impact on pre-final and final effluent sBOD/BOD with and without Memphis Cellulose being on-line. The impact of Runyon on pre-disinfection color is not obvious when Memphis Cellulose is digesting but is obvious when Memphis Cellulose is not digesting. The impact of Runyon on Pre-disinfection effluent quality in isolation can be ascertained as follows:

	Influent Concentration Increase with Runyon on line and Memphis Cellulose not digesting	Pre-final Concentration Increase with Runyon on line and Memphis Cellulose not digesting	Percent Pass Through to Disinfection for Runyon
COD	217	20	9%
Apparent Color	257	50	20%
True Color	129	40	31%

As shown above, as a percentage, more True Color contributed by Runyon potentially passed through to pre-disinfection than COD or Apparent Color. The impact of Runyon is masked by the fact that its discharge occurs over a single shift and not over a 24-hour period like the other industrial users and could be better described as a series of batch discharges.

The impact of all Other Users (excluding Memphis Cellulose and Runyon) on disinfection can be ascertained as shown in the following table for 2024:

	Influent Concentration for All Users Except Memphis Cellulose and Runyon	Pre-final Concentration Increase without Runyon and without Memphis Cellulose	Percent Pass Through to Disinfection for All Users Except Memphis Cellulose and Runyon
COD	713	131	18%
Apparent Color	2498	554	22%
True Color	378	193	51%

The impact of Runyon on final effluent quality under various scenarios is shown below:

	Memphis Cellulose Not Digesting		Memphis Cellulose Digesting	
Parameter	Runyon Off line	Runyon On line	Runyon Off line	Runyon On line
Final Eff BOD (adjusted for PAA impact)	23	37	30	37
Final Eff BOD (not adjusted for PAA impact)	29	46	38	49
Stiles Final Eff TSS conc	21	24	25	25
Days	67	130	39	129

The above table shows that the best effluent quality is achieved when Runyon is off-line and Memphis Cellulose is not digesting. The table also shows that Runyon discharge impact on effluent quality alone is similar when combined with Memphis Cellulose.

American Yeast has a 365/24/7 production schedule with minimal off time, so it is next to impossible to ascertain its statistical impact on disinfection unlike Memphis Cellulose and Runyon which have defined production schedules and off time. The impact of American Yeast is shown within the statistical conditions when Memphis Cellulose is not digesting and when Runyon is off-line. Note that Memphis Cellulose impact lingers for up to 5 days at the Stiles Facility disinfection facility as impact of color recycle diminishes.

	Memphis Cellulose Not Digesting and Runyon Off Line*
PAA Solution ppd	56,806
PAA dose mg/l	15.8
Stiles Final Eff PAA mg/l	1.2
Influent COD mg/l	713
Influent Apparent Color	2498
Influent True Color	378
Pre-final COD mg/l	131
Final Eff COD mg/l	112
Stiles Influent Flow mgd	82.7
E coli (99 percentile)	2134
E coli (geomean)	52
Number of E coli Daily Permit Exceedances	2
Stiles Pre-final Eff Color	554
Pre-final AIC 412 Eff Color	639
Pre-final True Color	193
Pre-final sBOD mg/l	14
Pre-final Eff BOD mg/l	29
Final Eff BOD mg/l	29
Increase in sBOD after disinfection	4.9
Final Eff Adj BOD mg/l	23
Days Running or Down	67

A summary of proportion of constituents in Stiles Facility influent for 2024 is presented as follows:

Constituent	All Others	Runyon	Memphis Cellulose	Total
COD	713	217	135	1065
Apparent Color	2498	257	-336	2755
True Color	378	129	438	945

A summary of the percentage of constituents in Stiles Facility influent for 2024 is presented as follows:

Constituent	All Others	Runyon	Memphis Cellulose
COD	67%	20%	13%
Apparent Color	91%	9%	None
True Color	40%	14%	46%

A summary of proportion of constituents in pre-disinfected effluent for 2024 as well as other information is presented below:

	All Other Users	Runyon	Memphis Cellulose	Total
COD	131	20	33	184
Apparent Color	554	50	194	798
AIC 412 Color	639	66	236	941
True Color	193	40	179	412

The above data is summarized on a percentage basis:

	All Other Users	Runyon	Memphis Cellulose
COD	72%	11%	18%
Apparent Color	69%	6%	24%
AIC 412 Color	68%	7%	25%
True Color	47%	10%	43%

The pass through of the influent constituents allocated to various users to pre-disinfection users is shown below:

	COD mg/l	True Color	Apparent Color
Influent-Memphis Cellulose	135	438	0
Pre-disinfection effluent-Memphis Cellulose	33	179	194
Pass Through-Memphis Cellulose	26%	41%	NA
Influent-Runyon*	217	129	257
Pre-disinfection Effluent-Runyon*	20	40	50
Pass Through-Runyon*	9%	31%	19%
Influent-All Other Users	713	378	2498
Pre-disinfection Effluent-All Other Users	131	193	554
Pass Through-All Other Users	18%	51%	22%

Assuming all three parameters have utility in allocating impact on PAA disinfection, True Color appears to be the most reliable and less prone to analytical interference due to the filtration step. It is possible that Soluble COD would produce similar results although that analysis was not performed. On average, the removal rate for Memphis Cellulose True Color is 59%. The removal rate for Runyon True Color is 69%. The removal rate for all others including American Yeast True Color is 49%.

Assuming American Yeast discharge comprises 50% of all other users True Color contribution in both influent and pre-disinfection, the three major color contributors (Memphis Cellulose, American Yeast, and Runyon) comprise 71% of the average influent True Color received by the Stiles Facility and 77% of the pre-disinfection effluent True Color. Memphis Cellulose comprises 43% of the pre-disinfection effluent True Color followed by American Yeast with an estimated 23% and Runyon with 10%. The percentage of calculated major True Color contributors is as follows:

	Influent True Color	Pre-disinfection Effluent True Color
Memphis Cellulose	46%	43%
Runyon	14%	10%
American Yeast (estimated)	20%	23%
All Other Users	20%	24%

The percentage of calculated major COD contributors is as follows:

	Influent COD	Pre-disinfection Effluent COD
Memphis Cellulose	13%	18%
Runyon	20%	11%
American Yeast (estimated)	33%	36%
All Other Users	34%	36%

The percentage of calculated major Apparent Color contributors is as follows:

	Influent Color	Pre-disinfection Effluent Color
Memphis Cellulose	Zero*	24%
Runyon	9%	6%
American Yeast (estimated)	41%	35%
All other Users	40%	34%

*The percentage of Apparent Color contributed by Memphis Cellulose in Stiles Facility influent could not be differentiated but likely is similar proportion to pre-disinfection effluent.

The relative proportion and percentage for PAA dosing and along with proportional disinfection costs is presented as follows:

	PAA Solution ppd	Percentage	Disinfection Costs (50%)
Memphis Cellulose	12,014	16%	$1,351,574
Runyon	7063	9%	$760,261
American Yeast (estimated)	28,403	37%	$3,125,516
All other Ius	28,403	37%	$3,125,516

The relative proportion and percentage for discharged average daily COD ppd as reported on Self-Monitoring Reports along with proportional COD surcharge revenue is presented as follows:

	Discharged COD ppd	Percentage	COD Surcharge Revenue
Memphis Cellulose	44,710	16%	$680,182
Runyon	89,570	30%	$1,275,341
American Yeast (estimated)	89,288	30%	$1,275,341
All other IUs	71,914	24%	$1,020,273

Finally, relative proportion of disinfection costs and COD surcharge is presented as follows:

	Disinfection Costs (50% of total)	COD Surcharge Revenue	Percent Disinfection Costs covered by COD Surcharge
Memphis Cellulose	$1,351,574	$680,182	50%
Runyon	$760,261	$1,275,341	168%
American Yeast (estimated)	$3,125,516	$1,275,341	41%
All other IUs	$3,125,516	$1,020,273	33%

Note that the COD Surcharge was based on the industrial COD discharge and projected disinfection cost for both Maxson and Stiles Facilities combined.

Internal Metals Investigation

The Stiles Facility had a process upset late 2023 and early 2024. Elevated metals were shown to be in the activated sludge and therefore were potential contributors to the process upset. An investigation was performed looking at external and internal sources of metals.

Grab samples were collected of the influent (one by City and another by Tioga), grit tank influent, plant final effluent, RAS A and B, Dewatering Filtrate, Lagoon Supernatant, and Biosolids cake starting 4-5-2024 through 1-23-2025 and analyzed for Metals. Metals concentrations were converted to mass values based on typical flowrate for each sample location. The mass values are provided in the following table:

Location	Chromium ppd	Copper ppd	Nickel ppd	Zinc ppd
Influent-City	4.14	41.33	13.56	142.11
Influent-Tioga	4.09	52.46	14.66	148.92
Influent Grit	6.1	60.31	17.2	190.67
Effluent	0	4.08	5.44	17.18
Wasting to Lagoon	3.04	40.28	6.6	106.84
Dewatering Filtrate	0.41	4.59	1.16	14.24
Lagoon Supernatant	0.27	3.2	1.21	9.7
Biosolids	2.4	22.8	6.2	64

If you perform a mass balance using influent as the input and effluent and waste activated sludge as the outputs, the following accumulation rate within the basins and channels is shown:

Location	Chromium ppd	Copper ppd	Nickel ppd	Zinc ppd
Basin/Channel Accumulation based on Influent	1.08	2.54	2.07	21.50

This condition is evidenced by the tremendous quantity of solids/metals that were removed from the final and return channels during a clean-out performed in September 2024 for Side B. The final channel was 40% filled and the return channel was 65% filled with accumulated grit and

other solids for a combined weight of 1.75 million wet pounds removed. It is noted that a lot of material remained in the channels. The quantity of chromium, copper, lead, nickel, and zinc removed from the Side B channels was determined as follows:

Constituent	Pounds
Chromium	91.2
Copper	403.7
Lead	164.2
Nickel	138.8
Zinc	629.8

Metals in the influent grit tank are consistently at higher levels than in the plant influent. The influent grit sample includes drainage from final tanks and aeration basins and other downstream areas. It is believed that internal recycling is occurring as well as on-going accumulation since the only exit points for these metals would be final effluent and waste activated sludge. The internal recycling for metals is shown in the following table:

Location	Chromium ppd	Copper ppd	Nickel ppd	Zinc ppd
Basin/Channel Grit Recycling	3.06	15.95	5.16	66.65

A mass balance was performed of the two covered lagoons with waste activated sludge as an input and dewatered biosolids, dewatering filtrate, and lagoon supernatant as outputs. The following accumulation rates within the two covered lagoons are shown:

Location	Chromium ppd	Copper ppd	Nickel ppd	Zinc ppd
Lagoon Accumulation	-0.04	9.69	-1.97	18.9

The return activated sludge was sampled weekly for metals. There was a downward trend for chromium and lead with a low point occurring soon after the RAS B channel cleanout followed by a recovery. Copper levels were steady although they took a temporary dip after the Side B channel cleanout. Nickel has been on a downward trend likely because of negative accumulation in the covered lagoons. Like copper, zinc levels are steady although they took a temporary dip after the Side B channel cleanout.

External Metals Investigation

The Stiles Facility had a process upset late 2023 and early 2024. Elevated metals were shown to be in the activated sludge and therefore were potential contributors to the process upset. An investigation was performed looking at external and internal sources of metals. The results of the investigation pertaining to external source of metals was in four phases and is as follows:

Phase 1

Grab samples were collected of the influent (one by City and another by Tioga), dewatering filtrate, lagoon supernatant, Loosahatchie Interceptor, Mud Island Interceptor, and Wolf River Interceptor on a weekly basis starting 4-5-2024 through 1-6-2025 and analyzed for metals.

Metals concentrations were converted to mass values based on typical flowrate for each sample location. In addition, IMPT provided metal mass values from industry self-monitoring reports and/or control authority results from the approximate first 9 months of 2024. The mass values are provided in the following table:

Location	Chromium ppd	Copper ppd	Nickel ppd	Zinc ppd
Dewatering Filtrate	0.41	4.59	1.16	14.24
Lagoon Supernatant	0.27	3.2	1.21	9.7
Loosahatchie Int.	1.91	3.53	2.09	23.09
Wolf River Int.	0	21.77	8.29	52.92
Mud Island Int.	0	2.36	0	2.87
AY	0.98	0.501	NA	12
KTG	0.063	0.116	NA	2.24
Van Gold/Pear Street	NA	NA	0.818	0.19
Influent-City	4.14	41.33	13.56	142.11
Influent-Tioga	4.09	52.46	14.66	148.92
Percent Explained-City	88%	87%	100%	83%
Percent Explained-Tioga	89%	69%	93%	79%

Based on the above results, more SMR data is needed for AY, KTG, and Van Gold for chromium, copper, and zinc. Note that there could be other industries on Pear Street contributing metals.

Phase 2

A mass balance was performed on the Loosachatchie Interceptor sampling location comparing results with Runyon and Corovo Mining, both upstream. The mass values are provided in the following table:

Location	Chromium ppd	Copper ppd	Nickel ppd	Zinc ppd
Corovo Mining	1.55	4.62	1.2	1.54
Runyon	0.348	NA	0.474	9.62
Loosahatchie Int.	1.91	3.53	2.09	23.09
Percent Explained	99%	131%	80%	48%

The mass balance for Chromium and Copper is nearly 100% or greater. The mass balance for Nickel is 80% and for Zinc is 48%. This indicates that the two industries could be underreporting for Nickel and Zinc or there may be other industrial user(s) not accounted for.

Phase 3

A mass balance was performed on the Wolf River Interceptor sampling location comparing results with Memphis Cellulose, PMC Biogenics, and Wolf River Interceptor manhole near Warford which is upstream. The mass values are provided in the following table:

Location	Chromium ppd	Copper ppd	Nickel ppd	Zinc ppd
Memphis Cellulose	NA	1.95	0.345	3.37
PMC Biogenics	NA	NA	5.34	8.34
Wolf River Int. @Warford	0	21.16	16.32	42.33
Wolf River Int.	0	21.77	8.29	52.92
Percent Explained	NA	106%	265%	102%

Chromium levels were very low so no mass balance could be performed. The Copper and Zinc mass balances were perfect at 106% and 102% respectfully. The Nickel mass balance blew up with high nickel values just at the Wolf River Interceptor @Warford location that were not showing up further downstream. The reason for this could not be explained.

Phase 4

A mass balance was performed at the Wolf River Interceptor @Warford location comparing results with PennaChem and Wolf River Interceptor @Highland which is upstream of where PennaChem comes in. The mass values are provided below:

Location	Chromium ppd	Copper ppd	Nickel ppd	Zinc ppd
PennaChem	0.028	3.82	1.09	0.28
Wolf River Int. @Highland	1.91	9.82	1.18	19.62
Wolf River Int. @Warford	1.28	21.16	16.32	42.33
Percent Explained	151%	64%	14%	47%

It was tough getting good mass balance at this location. There appears to be Chromium in the vicinity that is not showing up closer to the treatment plant. PennaChem could be underreporting for Copper, Nickel, and Zinc. At the same time, there could be another industrial user in this general area who is not reporting Copper, Nickel, and Zinc. There could also be a significant source of Zinc upstream of Highland.

Summary

The following is the percent of Chromium, Copper, Nickel, and Zinc in the Stiles Facility influent:

Location	Chromium ppd	Copper ppd	Nickel ppd	Zinc ppd
Percent Side Streams	17%	17%	17%	16%
Percent known industrial	72%	23%	66%	26%
Percent Mud Island Interceptor	0%	5%	0%	2%
Percent Unexplained from Looshatchie Int.	0%	0%	3%	8%
Percent Unexplained from Wolf River Int.*	0%	34%	11%	28%
Percent Unexplained from general vicinity of Stiles Facility**	12%	22%	4%	19%

* possible underreporting by Memphis Cellulose, PMC Biogenics, PennaChem or other sources not yet identified
** possible underreporting by AY, KTG, Van Gold or other sources on Pear Street

Another way to summarize the data is to trend the metals loading (ppd) along the Wolf River Interceptor going from west to east, ending at Highland Street. The results are shown as follows:

Location	Chromium	Copper	Nickel	Zinc
Influent Wet Well	4.14	41.33	13.56	142.11
Wolf River Interceptor@North 2nd	0	21.77	8.29	52.92
Wolf River Interceptor@Warford	0	21.16	16.32	42.33
Wolf River Interceptor@Highland	1.92	9.82	1.18	19.62

The data shows the increase in metals loading associated with industries between North Second Street and Highland Street as well as sample variability associated with chromium and nickel. A significant quantity of metals appears to be originating between Highland and Warford. East of Highland would consist of both residential and commercial users as well as a few industrial users located in Barlett.

COD Surcharge versus Disinfection Costs

The 2024 COD surcharge for the industries that the Stiles Facility serves generated an estimated $4,251,138 in revenue. The 2024 disinfection cost was $16,894,680. The COD surcharge is expected to cover 50% of disinfection costs or $8,447,340. This is a shortfall of $4,196,202 or 50%.

Biogas Investigation/Cogeneration

51 samples of biogas were collected throughout the year and analyzed for siloxanes, methane, carbon dioxide, oxygen, nitrogen, hydrogen sulfide, and for BTU. Total Siloxanes as silicon averaged 54 ppbv ranging from 16 to 106. Hydrogen Sulfide concentrations averaged 5,054 ppmv ranging from 350 to 21,000. Energy values averaged 470 BTU/cubic feet (wet) ranging from 160 to 630.

Year	Soloxanes (ppbv)	Hydrogen Sulfide (ppmv)	BTUs/cubic foot (wet)
2013	136	7,331	639
2014	107	4,023	602
2015	NA	3,129	570
2016	159	3,266	559
2017	131	3,083	638
2018	72	4,150	591
2019	17	3,554	588
2020	28	3,850	568
2021	57.6	9,500	464
2022	74.5	4,240	381
2023	78.7	3,609	484
2024	54	5,054	470

The previous Stiles Facility Air Permit effective 9-16-2019 through 9-16-2024 was received on 10-2-2019. The City received a Notice of Inquiry on 5-2-2021 for not increasing air permit sampling when SO2 emissions exceeded 50 tpy on a rolling 12-month basis. The City met with SCHD to review the situation on 6-4-2021 and it was agreed that the City would upgrade to T5 permit. The City provided comments to the draft new T5 permit on 9-22-2021. A new T5 permit was issued 3-1-2022 to expire 3-1-2027. A new permit application would need to be submitted no later than about 9-1-2026.

The Shelby County Health Department granted an extension of a construction permit through 8-25-2022 to allow the connection of a natural gas line to the co-gen unit. This will also involve re-commissioning the co-gen for service and then firing it with natural gas while projects are implemented to both increase biogas production and remove hydrogen sulfide. A kick-off meeting was held with Malasri Engineering on 4-14-2021 to convert the co-gen from biogas to natural gas. The Stiles Facility worked with Malasri Engineering to run natural gas to the co-gen in February 2022. A permit application for a new construction permit for natural gas fired co-gen was submitted 11-14-2022. SCHD issued the construction permit effective 3-15-2023, expiring 9-15-2024. On 8-28-2023, the new natural gas meter was finally connected to the Stiles Facility co-gen system after months of starts and stops. On 1-18-2024, the Stiles Facility co-gen was able to run on natural gas for the first time and continued to be fired monthly for the remainder of the year as part of on-going maintenance agreement with Cummins pending receipt of new construction permit. A new construction permit application was submitted toward the end of 2024.

The Semi-Annual Report covering the July-December 2023 period as required by the T5 permit was submitted to SCHD on 2-16-2023. SCHD conducted their annual inspection on 3-21-2024. The Semi-Annual Report covering the January-June 2024 period was submitted on 7-26-2025,

<u>Pre-Final and Final Effluent Data Comparison—The Golden Goose</u>
Beginning late 2017, sampling has been performed upstream (Pre-final) and downstream (Final) of the secondary treatment effluent disinfection basin. The following table presents average concentrations of samples collected:

Sample ID	Analysis	2017 Ave	2018 Ave (before disinfection started)	2018 Ave (after disinfection started)	2019 Ave	2020 Ave	2021 Ave	2022 Ave	2023 Ave	2024 Ave
Pre-Final Effluent	BOD	43	50	44	70	42	41	27	29	41
Pre-Final Effluent	sBOD		28	20	36	13	18	14	15	22
Pre-Final Effluent	COD				256	102	150	125	93	159
Pre-Final	TSS	49	56	46	73	57	72	30	29	45

Effluent										
Pre-Final Effluent	Apparent Color*			815	822	460	571	539	509	679
Final Effluent	BOD	30	33	42	59	26	33	26	26	43
Final Effluent	sBOD		23	23	43	20	26	19	21	30
Final Effluent	COD	177	197	208	238	97	123	110	86	149
Final Effluent	TSS	29	31	25	40	16	15	9	11	24

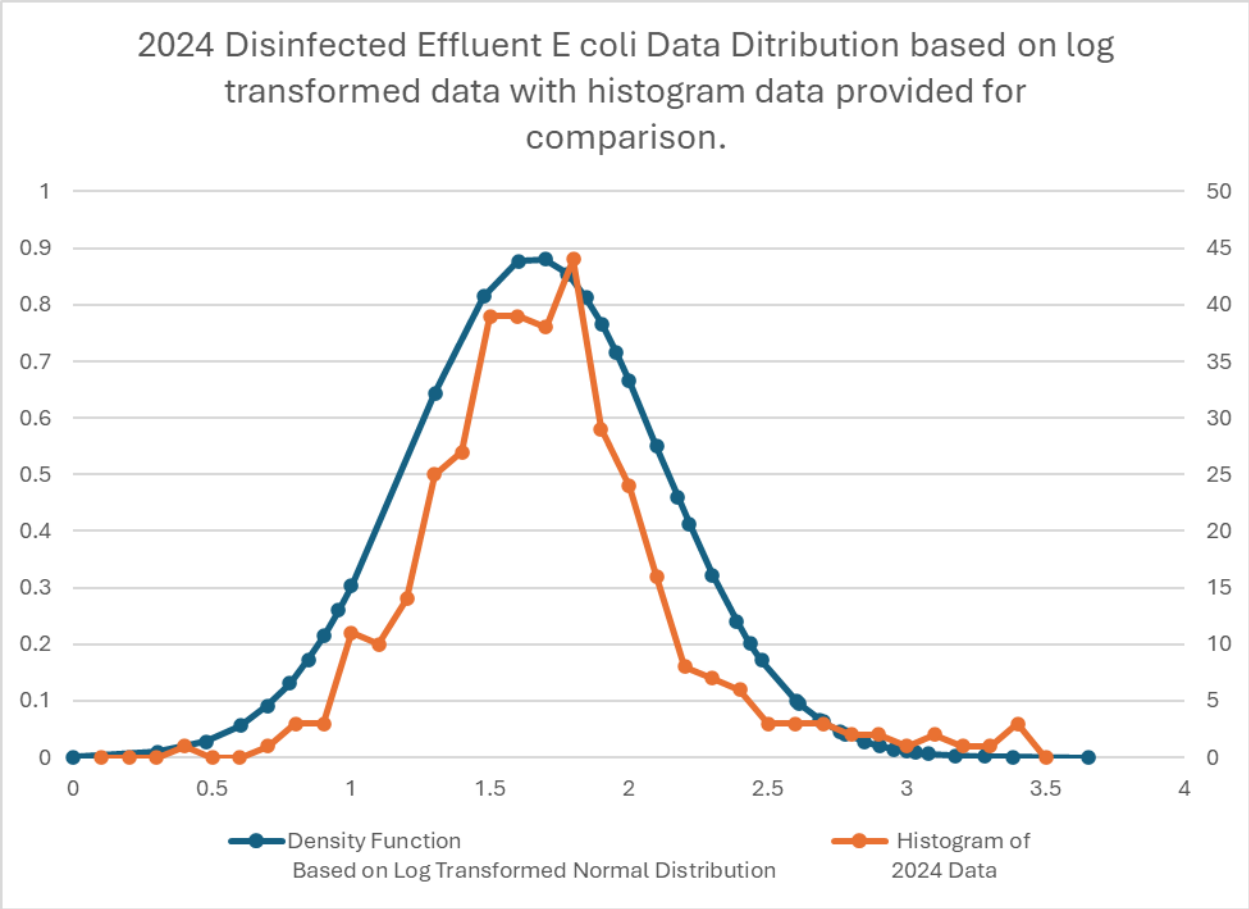
*based on City Lab Color

The data gives an indication of the ability of the effluent disinfection tank to provide additional, post-secondary removal of pollutants (i.e. The Golden Goose). Periodic disinfection began on 9-12-2018 and full-time disinfection began on 11-10-2018. After 9-12-2018, Final Effluent soluble BOD is higher than Pre-Final Effluent soluble BOD. Prior to 9-12-2018, Final Effluent soluble BOD was consistently lower than Pre-Final Effluent soluble BOD. Initially, it appeared the installation of the vertical baffles to improve mixing of the injected PAA had potentially increased TSS removal after the high flow year of 2019. However, a significant portion of the decrease is now believed due to improvements in solids recovery implemented by Memphis Cellulose after 2019. One mystery presented here is why Pre-final Effluent sBOD in 2018 was higher than Final Effluent sBOD prior to startup of disinfection.

Regarding the Golden Goose, in 2024, the pre-final effluent BOD average was 41 mg/l and the final effluent BOD average was 43. The pre-final sBOD was 22 and the final sBOD was 30 and the pre-final TSS was 45 and the final TSS was 24. Therefore, the increase in sBOD was somewhat offset by the decrease in TSS.

E Coli Statistical Review

Below is the actual population curve based on the 2024 data:



The following statistics are derived for the entire year:

Result	Percentile
46	50%
126	83.5%
244	94.6%
410	98.3%
487	98.9%

Based on the above table, there would be 4 daily max violations based on 487 daily max and 6 daily max violations based on 410 based on the hypothetical population curve. There were 12 actual daily max violations based on 487 daily max many of which were due to surface conditions causing sample interference. Based on the above table, there were 20 occurrences above the contract daily limit. of 244 based on the hypothetical population curve and 24 actual daily contract limit violations.

Laboratory

The laboratory located at the M. C. Stiles Facility received samples from the Stiles Facility and the Maxson Facility. The laboratory performed a total of 30,781 analyses, including 25,136 in-house which included 2163 E. coli tests and is compared to past years:

Year	2024	2023	2022	2021	2020	2019	2018	2017	2016	2015
Stiles	NA	NA	NA	NA	9355	12,953	13,019	10,089	13,157	11,775
Maxson	NA	NA	NA	NA	7734	12,167	11.931	15,121	22,537	18,952
Chapel Hill	NA	0	0	NA	578	603	559	549	533	447
Industrial Monitoring	0	0	0	0	0	2	0	0	339	617
Total	30,781	29,204	26,495	18,073	17,667	25,725	25,509	25,759	36,566	31,795

3.0 Analysis and Future Outlook

Analysis

On 1-10-1967, the City Commission adopted a resolution urging the State of Tennessee to participate in a sewage treatment program under the upcoming Federal Clean Water Act of 1968 for the construction of sewage treatment facilities. Sewer and industrial waste sampling commenced in 1967 as part of 201 Facilities planning under MATCOG and the US EPA. A flat monthly sewer charge for residential and other users was adopted by the City in July 1969 to provide local funding for these anticipated projects. On 2-5-1969, the Memphis Water Pollution Advisory Committee issued a statement that Memphis industries may have to shut down unless the City moves to clean-up its "open ditch" sewer system. On 4-8-1969, the City Council considered grants for the construction of two major wastewater treatment facilities as well as a new interceptor sewer for the Nonconnah Creek Basin. On 1-6-1970, the City Council adopted a resolution setting a construction schedule for secondary treatment facilities with the south plant to be completed by July 1974 and the north plant completed by December 1978. Contracts were negotiated with two joint-venture engineering teams to design and oversee construction of each treatment plant. Allen & Hoshall and Clark, Dietz & Associates were selected to design and oversee the construction of the south plant. Ellers, Reaves, Fanning, and Oakley and Consoer Townsend were selected for the north plant. Both design teams embraced contact-stabilization to meet the requirement for secondary treatment without the use of primary clarifiers as well as chlorine disinfection. The design, start up, and operation of each plant, however, followed different tracks and had different challenges.

The Maxson Facility treatment capacity history is summarized as follows:

Year	Flow (mgd)	BOD (ppd)	SS (ppd)
1975	80	186,000	133,500
1985	80	390,000	335,000
1997	90	490,000	410,000

The City purchased the land on 10-8-1971 for $65,000 from Shelby County to construct the Maxson Facility. In 1972, with the passage of P.L. 92-500, nationwide 30-30 effluent discharge limits were established for publicly owned treatment works constructed and in operation before 7-1-1977. The Maxson Facility design criteria pre-dated the 1972 legislation and only required 85% removal which would come to haunt the treatment plant later on.

The Maxson Facility was issued NPDES permit effective 11-30-1974 and expiring 12-31-1979. For the first two years of the permit, percent removal limits of 85% for BOD and TSS would be applied. Starting 1-1-1977, the 30-30 effluent concentration limitations would be applied. The plant officially opened on 6-4-1975 to much fanfare after having been in operation for about a week. Records from November 1975 through June 1977 indicate influent monthly average loadings approaching 300,000 ppd for BOD and 250,000 ppd for TSS and monthly average removals of 80% for BOD and 62% for TSS.

On 7-22-1977, the U.S. DOJ/EPA filed suit against the City for violations of the discharge permit from 6-30-1975 through 7-22-1977. The Court entered the Consent Decree on 7-28-1978 to increase plant capacity. The Decree was modified on three subsequent occasions: Amended Decree (2-23-1979); Second Amendment to Decree (5-7-1979), and Consent Order (12-31-1981). The work was performed utilizing ten contracts consisting of additional blower purchase (Blower #6), piping modifications, blower installation, electrical modifications, site preparation, three (3) primary clarifiers, administration building modifications, electrical reliability modifications, computer modifications, and secondary treatment modifications (4 ABF Towers, 5 Bio-pumps). The Decree was terminated on 10-10-1986.

The City elected to make the required improvements using EPA Construction Grants. The City received approval for its grant application from the EPA on 12-20-1979. The City entered into a Patent License and Consulting Services Agreement with Neptune Microfloc, Inc. in 1981 for the rights to use the activated bio-filter (ABF) process. The City obtained an Innovative Technology Grant on 3-8-1982 to utilize the ABF process as the first stage secondary treatment of a high strength (600 mg/l BOD5) wastewater. The EPA approved the funding of 4 instead of 6 bio-filters as originally requested taking the stance there was sufficient capacity through 1985 loading projections provided the bio-filters were fully loaded.

The Maxson Facility went on-line in March of 1986 with all 4 bio-filters in service. An Evaluation of Operating Data was performed on data collected from May 1985 to February 1986. The report showed that the plant would have trouble staying under 30 mg/l for effluent BOD and TSS during the colder months. On 10-24-1986, the City proposed a Supplement to Evaluation of Operating Data from the T.E. Maxson Wastewater Treatment Plant to the TDHE Division of Construction Grants. One bio-tower would be taken off-line to allow the other three to be fully loaded and then operated during the coldest time of the year. The evaluation was performed from December 1986 through February 1987. During this period, three bio-filters were operated with 4 bio-pumps, three bio-filters were operated with 3 bio-pumps, and two bio-filters were operated with 3 bio-pumps.

A review of operating data from December 1986 to February 1987 confirmed that the organic removal across the towers is a function of the transfer capability within the bio-towers which depends on the temperature of the wastewater and the hydraulic loading rate. The evaluation concluded that at maximum month design loads, the biotreatment system would fail completely and EPA effluent limitations would be exceeded. Among other alternatives, the report recommended adding two additional bio-pumps to increase the overflow rate and converting the 4 existing aerobic digestion tanks to regular aeration.

A letter from TDHE Construction Grants and Loans dated 2-29-1988 to the USEPA indicated the City had sent a modification request respective to an innovative design application and concluded that the upgrade would not consistently operate as designed. TDHE was transitioning to the State Revolving Fund program and existing M/R funds were dissipated so they requested EPA review the request in a timely manner. The M/R Grant was approved, and the construction performed 1989-1990.

The Maxson Facility existing primary clarifier sweeps were modified during 1991-93 to increase sludge removal. A second capacity expansion took place 1995-1997 consisting of the addition of two ABF Towers, a 4th Primary Clarifier/Thickener, and two Secondary Clarifiers. A covered lagoon capacity expansion project was completed in 2004 which covered Lagoons 2A, 2B, and 2C and improved solids capture and primary sludge pumping. Primary sludge dewatering was

discontinued in 2004 and the wasting of primary sludge directly to the lagoons for co-digestion with waste activated sludge began. In 2006, the bio-towers began periodic operation in roughing mode resulting in energy savings, improved effluent quality, and a dramatic reduction in waste activated sludge. This practice continued sporadically through the second half of 2019 when discontinued due to valve problems. In 2010, a new cover was put on Lagoon No. 1 and the influent head works replaced. In 2013, new Fine Bar Screens were put into operation, reducing plastic items wasted to the lagoons. The Primary Scum Removal System was reworked in 2014 increasing the overall removal efficiency of the primary clarifiers.

Co-digestion of primary sludge with waste activated sludge beginning in 2004 along with covered lagoon expansion led to a dramatic increase in biogas production that continued through 2016. In 2017, biogas production began falling off. There had been a decrease in activated sludge wasting in recent years due to increased use of roughing mode contributing to the decline in biogas production. In addition, inert solids build up in Lagoon 2A may have accelerated after the TVA ash release in 2010 contributing to the decline in biogas production.

The expected unmonitored influent BOD should be 69,000 ppd based on population size (an estimated 404,570 people) or 92,000 ppd based on minimum flow and 250 mg/l BOD. Figure 15 shows trends in unmonitored influent BOD. The increase in BOD at the head of the plant was totally unexpected after completion of major lagoon improvement projects which focused on solids retention.

Similarly, the expected unmonitored influent TSS should be either 81,000 ppd or 110,000 ppd based on minimum flow and 300 mg/l TSS. Figure 16 shows trends in unmonitored TSS. The monitored result includes the 24-inch dewatering filtrate / lagoon supernatant sampling point. Two other return flow lines including Lagoon No. 1 and Horn Lake Ditch are not monitored. The graph shows major spikes in unmonitored influent TSS occurring since 2004. The cause of these spikes can be attributed to unmonitored lagoon returns and unrepresentative influent sample collection.

Average annual effluent BOD concentration since 1975 is shown in Figure 17. The figure shows gradual increase related to increases in influent loading for a 20-year period beginning in 1987. For the last 10 years or so, increases appear to be due to worsening removal efficiency.

Average annual effluent TSS concentration since 1975 is shown in Figure 18. The figure shows gradual increase related to increases in influent loading beginning in 1987. The increases over the last 10 years or so appear to be due to worsening removal efficiency.

As shown in Figure 19, combined primary and secondary sludge wasting peaked in 2006. Beginning that year, as previously discussed, the bio-towers began to be periodically operated in roughing mode for the first time producing less waste activated sludge. The Maxson Facility has been operated mainly in ABF mode since after mid-2019. As shown in Figure 20, annual power consumption peaked between 1997 and 2004. Power consumption began decreasing in 2005 as the plant began wasting primary sludge directly to the lagoons and operating more in roughing mode beginning 2007 resulting in greater efficiency of the bio-towers and less blower operation as well as improved effluent quality. This downward trend continued until 2010. Energy consumption has been increasing since then due to inability to maintain consistent roughing mode among other conditions that tend to increase blower usage despite consistent influent loading.

The City contracted with the joint venture for engineering services on 10-31-1969 to begin design of the Stiles Facility originally known as the North Sewage Treatment Plant. The City acquired the site location in 1971 consisting of 208 acres, including about 86 acres located on land fill produced by the spoil of the ACOE dredging work in stabilizing the Mississippi River and Wolf River channels in 1965. The surface elevation of the land fill area where the main plant facility is situated varies 240 to 242 feet AMSL. Prior to dredging, the natural surface area elevation varied 210 to 225 feet AMSL. Approximately 20 acres of additional land fill surface area adjacent the Mississippi River was deeded to the Lucy-Woodstock Utility District for barge docking and chemical supply station for an industrial pipeline. The area immediately north of the main plant facility was acquired by the Airport Authority for relocation of a small municipal airport formerly located on the southern end of Mud Island. The airport was constructed and opened to the public on 5-30-1971. A report "Basic Design Data for Preparation of Contract Drawings" was issued June 1971, contained the basic parameters for plant operations. The Stiles Facility was issued NPDES permit effective 11-30-1974 and expiring 12-31-1979. For the first two years of the permit, percent removal limits of 85% for BOD and TSS would be applied. Starting 1-1-1977, the 30-30 effluent concentration limitations would also be applied.

The Stiles Facility began operation in August 1977. By January 1979, the plant was still not consistently meeting its 30-30 effluent concentration limitations. CH2MHill was hired to evaluate operating data and to determine the maximum treatment capabilities of the plant in its present configuration at existing and future loading rates. The report issued April 1980 concluded that major plant modifications would be required (i.e. adding primary clarifiers) if future influent loading conditions (i.e. 450,000 ppd BOD) increased above its current design load (275,000 ppd BOD) assuming that aerobic digestion was eliminated. Full scale operational testing showed that the plant could meet 30-30 standards at design loading by operating 8 contact and 10 stabilization tanks. It was further recommended that the permit limits be raised to 60-60 due to the high industrial contribution to the plant.

As discussed, the design, start-up and operation of the Stiles Facility has been relatively straight forward with no major process modifications since original construction. Its original design reportedly was based on the Stinkney Plant in Chicago. The aerobic digestors were discontinued in the early 1990s. Industries have come online in recent years to bring the plant up to full treatment capacity. In 2003, American Yeast began production in February and KTG began production in March. Discharges increased as these facilities ramped up to full production. In 2013, KTG expanded production with the startup of its TAD unit. Runyon's evident expansion of operations in 2021 in an apparent sneak attack pretty much topped off the Stiles Facility BOD tank along with other refractory qualities.

Trends in unmonitored influent BOD loading are shown in Figure 21. The expected unmonitored influent BOD should be 73,000 ppd based on population size (an estimated 427,435 people) or 85,000 ppd based on minimum actual flow and 250 mg/l BOD. Until the last couple of years, the lowering and increased stability of that number would indicate that industrial discharges are being effectively monitored for that parameter. However, unmonitored BOD has been increasing to a level of concern. Unmonitored influent TSS has historically been erratic and higher than expected over the years but has been becoming steadier and trending upward as shown in Figure 22. The expected unmonitored influent TSS based on population this size (an estimated 427,435 people) is 85,000 ppd or 103,000 ppd based on minimum actual flow and 300 mg/l TSS.

Average annual effluent BOD concentration since 1978 is shown in Figure 23. Since the plant began operation in 1978, effluent BOD concentration has been a function of influent loading until after 2019 when there was a sudden decrease in concentration apparently related to Memphis Cellulose installing a fiber recovery unit.

Average annual effluent TSS concentration since 1978 is shown in Figure 24. Over the last 20+ years, effluent TSS concentration has been a function of influent loading with the relationship being one of general trend direction and not one-to-one.

As shown in Figure 25, sludge wasting peaked in 1988 and was subsequently on the decline for many years reaching a low point in 2003 when both American Yeast and KTG both began production. Sludge production then picked up mirroring the discharges of these two industries. Presumably, American Yeast's high strength organics was converted to solids and wasted from the activated sludge process and KTG's mainly very low volatile solids were direct wasted to the lagoons. After the economic slowdown, sludge wasting began picking back up in 2014, helped by the increase in influent solids loading after the addition of the TAD unit at KTG. As shown in Figure 26, energy consumption was on a 20-year decline starting in 1981 as the plant implemented the Aeropt system and discontinued aerobic digesters. Energy consumption began increasing in 2003 when both American Yeast and KTG began production. Aeropt was discontinued in 2010 and energy consumption remained flat for several years. Energy consumption has increased in recent years due to increased blower usage to improve effluent quality, especially upstream of PAA addition and deterioration evident in secondary clarifier weirs. The decrease in energy consumption in 2023 was caused by blower problems brought on by summer thunderstorms.

The cost that MLGW charges for electricity was in the news for a while with talk about MLGW severing ties with TVA. Essentially, the wastewater treatment plants pay a wholesale rate from TVA. The average annual all-in cost for main power going back to 1996 is shown in Figure 27. The 2024 wholesale rate for power by the City's wastewater treatment plants was 8.15 cents per kwh. Stripping away other costs, the City is billed approximately 3.1 cents per kwh for power generation. The MISO wholesale rate for power generation in 2024 was around 3.1 cents indicating the City is getting a fair deal from TVA. This comparison is made on the assumption that the City of Memphis would be charged a commercial rate if power was furnished by MISO instead of the wholesale rate charged by TVA.

Table 1 and Table 2 both show influent concentration and loading and effluent concentration for BOD, COD, and TSS over the years. Table 3 compares total influent flow, average daily dry weather flow, and average daily billed wastewater flow at both treatment plants compared with drinking water production for each month. The total influent flow is the average daily flow received at the treatment plants. The average daily dry weather flow is based on days with no apparent river or rain impact. The average daily billed wastewater flow is based on residential and commercial water usage and billed large industrial user and wholesale customer sewer usage. Drinking water production is the metered output of the water treatment plants that is sold to residential and commercial customers and a small portion of it is not affiliated with the City. Several industrial users use private wells to generate cooling and process water instead of using MLGW drinking water and is discharged to the sanitary sewer on a continuous basis and may

constitute 10% or more of dry weather flow received. Figure 30 shows the Table 3 data over the last 10 years.

Table 4 and Table 5 present average annual rain-induced flow per inch of rain, percent of average daily flow that is rain-induced, average daily flow, average daily dry weather flow, annual rainfall, and percent of average daily flow that is river-induced for each treatment facility since 1975 and 1979 for the Maxson and Stiles Facilities, respectively. Trends in rates of rain-induced flow are shown in Figures 28 and 29. Over the last 10 years, rain-induced flow has averaged 28 million gallons per inch of rain at the Maxson Facility and 27 million gallons per inch of rain at the Stiles Facility. Trends in average daily and average daily dry weather flow for both facilities combined are shown in Figure 30. Overall, rain-induced flow has comprised 6% and 7% of average daily flow received at Maxson and Stiles, respectively, and river-induced flow has comprised zero % and 5% of average daily flow received, respectively. Combined ADF peaked in 1998, reached its lowest point in 2006, and climbed to a new peak in 2019. Combined ADDWF peaked in 1996 and was on steady decline through about 2016 and has been slightly increasing since then whereas billed daily flow as shown in Figure 31 has been steady for the last 10 years.

Data related to plant operations and sludge production is presented in Table 6 and Table 7. Some additional tables presented are Table 8, the monthly plant operating budget expenditures and gallons treated, and Table 9, a breakdown of plant expenditures by category. Treatment costs averaged 111 cents per 1000 gallons in 2023, a decrease of 8 cents from the previous year as shown in Figure 32 with previous years for comparison. Annual cost to treat infiltration / inflow are shown in Figure 33. These costs are more significant now that disinfection of secondary effluent has started. Table 10 and Table 11 comprise the largest industrial user's monthly fees for each plant. Table 12 shows the monthly fees of the three wholesale customers. These tables show total fees of $54,991,838 from these 19 industrial users, hauled wastes, and 3 wholesale customers. Fees are estimated this year. The estimated fees from the 19 industrial users, hauled wastes, and 3 wholesale customer fees covered 84% of the costs. The total estimated fees received increased $5,876,238 for the 19 industrial users and was assumed not changed for the wholesale customers for a total increase of $5,876,238 for these revenue categories from the previous year. More than likely, the industrial fees increased only slightly.

Table 13 is a monthly comparison of the utility charges for both treatment plants. The treatment plants changed to Time of Use contracts effective 10-1-2015 which makes unit cost tracking more complicated due to different on-peak and off-peak time blocks.

Table 14 shows the total loading of BOD_5 of the plant and the amount contributed by the major industries. Table 15 shows the total loading of COD of the plant and the amount contributed by major industries. Table 16 shows the total loading of total suspended solids of the plant and the amount contributed by the major industries. Table 17 shows the water usage and cost. Table 18 is a table showing the electrical usage, cost, biogas production, and propane usage for the covered lagoon. Table 19 is a table showing the electrical usage and cost for other electrical meters. The City used an average of 0.8 mgd of metered potable water at the Maxson Facility for cooling equipment and dewatering spray down and could be asked to transition to a lower quality water source based on future industrial need and potential new regulatory requirements. The actual usage is actually higher due to a meter malfunction beginning in August. Table 20 shows long-term trends in BOD loading and wasting.

Table 21 shows the utility usage, cost, and biogas production for the Stiles Facility. Table 22 shows the plant water usage and cost as well as the electrical usage and cost for dewatering. Table 23 shows total loading of BOD_5 for the Stiles Facility and the amount contributed to it by the major industries. Table 24 shows the total loading of COD of the plant and the amount contributed by major industries. Table 25 shows the total loading of TSS for the plant and the amount contributed by the major industries. The City used an average of 0.22 mgd of metred potable water at the Stiles Facility for cooling equipment and dewatering spray down and could be asked to transition to a lower quality water source based on future industrial need and potential new regulatory requirements. This average appears low and should be closer to 1 mgd and indicates possible problems with the meter. Biogas purchases by American Yeast are shown in Table 26.

Table 27 summarizes the electrical usage, total cost, unit charges, and savings over the past several years for both treatment facilities. Table 28 and Table 29 show comparisons of the Stiles Facility and Maxson Facility influent flows versus the combined flows of the major industries.

Recent Regulatory Developments

Final NPDES permits for both treatment plants were issued on 12-2-2011 for the Stiles Facility and on 12-6-2011 for the Maxson Facility, effective 1-1-2012 and expiring on 12-31-2016. Certain elements of the permits were appealed. TDEC proposed a modification to the Maxson Facility permit on 9-10-2012 based on comments received on the draft permit public noticed on 4-9-2009 and re-public noticed on 10-11-2010 as part of settlement negotiations related to the permit appeal. The proposed modified permit rationale recognized the variance from federal treatment standards, the deletion of mass limits for effluent parameters (like Stiles), and the deletion of effluent flow metering pending a flow study. However, TDEC did not pursue the matter further. Applications for new permits were submitted 7-5-2016. The City met with TDEC on 4-10-2017 to review the issuance of new permits. TDEC presented information at the meeting indicating the requested secondary treatment adjustments (i.e. variances) would not be granted. On 10-20-2017, the City submitted a report explaining why the variances should be granted as a supplement to the previously submitted permit applications. New draft permits for both treatment plants were public noticed on 11-13-2018. The permits stuck to the nationwide 30-30 standards and did not allow Best Professional Judgement (BPJ) to be used for industry groups without promulgated categorical effluent limits for secondary treatment adjustment. Public Works officials drove to Nashville on 12-10-2018 to meet with TDEC and learned that TDEC was not flexible on this issue. The City was granted an extension request for public comment to 2-15-2019. A public meeting for the Maxson and Stiles Facility proposed NPDES permits was held on 2-26-2019. TDEC would not impose a permit limit on PAA residual since the City would have an incentive to keep it low due to cost. The City submitted comments on 3-20-2019, totaling 800+ pages. On 4-18-2019, Vojin Janjic emailed that based on the comments received both NPDES permits would be withdrawn, and a new draft permit issued later which would require a new public notice. As the permit negotiations continued, in early 2021, TDEC legal personnel concluded that the state did not have the authority to implement secondary treatment adjustments without additional language in the regulations. On 4-20-2021, the Tennessee Water Quality Board approved the City's proposed amendments to the state regulations that would allow adjustments to TBELs in accordance with 40 CFR 133.103 (b). The amended regulations were promulgated on 5-15-2022. On 10-14-2022, the City submitted the most recent industrial user data in response to TDEC request related to Secondary Treatment Adjustments for the next NPDES permits. Previous NPDES permit limits for treatment and performance are

shown in Table 30. As shown in the table, secondary treatment adjustments of effluent limits were required for compliance and have varied, getting somewhat looser over the years.

The new NPDES permit was public noticed on 7-19-2023. On 8-17-2023, the City made a request to TDEC for a 90-day extension from 9-9-2023 to 12-7-2023 to provide NPDES permit comments.

The public hearing for the NPDES permits was held at the Main Library on 8-28-2023 with written comments due 12-7-2023 and final permits expected to be issued after Christmas. At the meeting, TDEC emphasized the permit limits being technology and not water quality based. There was also interest expressed by some in the audience regarding effluent re-use which TDEC indicated was covered by the regulations. The proposed versus current permits limits for BOD and TSS are shown below:

Maxson Facility	Monthly Average	Weekly Average	Daily Max
Proposed BOD	34.8	52.2	51.1
Current BOD	42	63	84
Proposed TSS	37.8	56.8	54
Current TSS	48	72	96

Stiles Facility	Monthly Average	Weekly Average	Daily Max
Proposed BOD	53.8	80.7	107.6
Current BOD	43.1	64.7	86.1
Proposed TSS	40.7	61.1	81.4
Current TSS	53.2	78	104

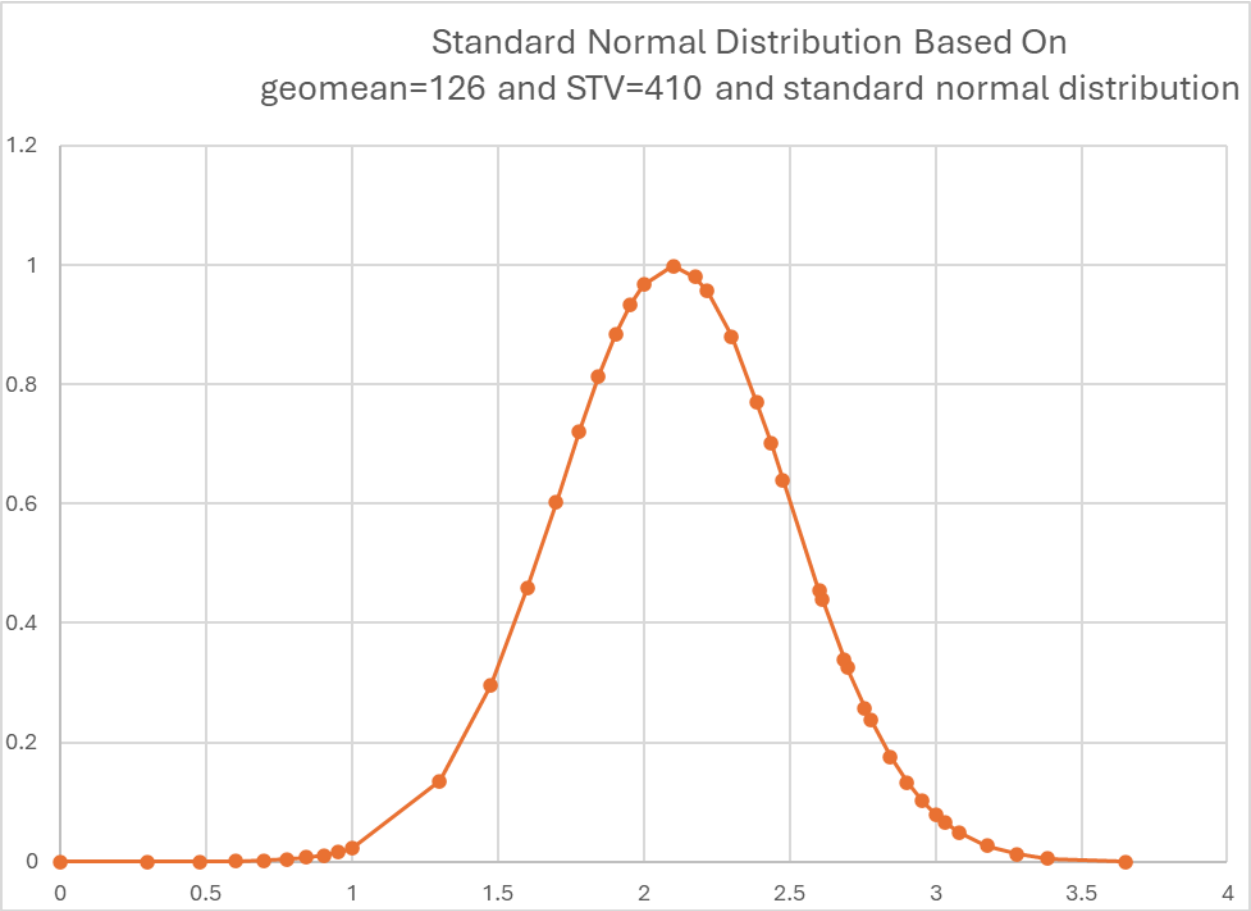
City comments were submitted on 12-7-2023. The main concern were the "rules" developed by the permit writer in the rationale for application of the federal secondary treatment adjustment regulations in the most stringent manner and the definition of reasonable measure of industrial production. The amended Uniform Administrative Procedures Act which was enacted in 2022 was cited stating that a court appeal would not consider a state agency's interpretation of the statute or rule.

On 3-17-2024, TCA Chapter 0400-40-40-03 General Water Quality Criteria amendments were enacted. The amendment, in part, included changes to the criteria for the use of Recreation as follows: "Additionally, the concentration of the E. coli group in any individual sample shall not exceed 410 cfu per 100 ml more often than in 10% of the samples during any 30-day period."
For monthly reporting, the E coli geomean would remain 126 cfu/100 mls. However, the daily max would be 410 cfu based on 10% daily excursions. The previous daily max was 487 cfu. When questioned about the 10% daily excursions, TDEC claims that the 10% daily excursions is "built in" and not an allowance and that the 410 cfu be looked at simply as a daily max and not in conjunction with a statistical population. The former 487 cfu standard was adopted after the 1997 Triennial Review as a single-sample criteria as part of the transition from Fecal Coliform to E coli WQS (i.e. E coli derived from equivalent Fecal Coliform). It does not appear that TDEC made this transition very clear.

EPA/FDA guidance, on the other hand, appears to suggest that the population curve for a 30-day period be calculated and the 90% statistical threshold value be calculated. This 90% threshold

value is then compared with the 410 cfu daily maximum.

The below graph shows the Standard Normal Distribution curve based on population geomean = 126 (log10 = 2.1) and standard deviation = 0.3996 based om Statistical Threshold Value = 90% or 410 (log10=2.612). Plotted on this same graph, the former 487 cfu daily maximum would have a STV of 92.9% or log10=2.6875.



City personnel met again with TDEC on 9-19-2024. TDEC's main concerns were the TNPOP, E coli, BOD-adjustment for PAA, and the 7 "rules" for secondary treatment adjustment. TDEC wanted to remove the "rules" section from the previous draft permit authored by Bob Odette showing how the federal secondary treatment adjustment standards were to be applied to the City of Memphis permits which appeared to be an attempt on the part of the permit writer to allow the permit to be challenged by the City for unauthorized rule-making and force TDEC to amend the regulations to include his ad hoc "7 rules". It is also likely that TDEC will need to amend the Water Quality Criteria regulations to eliminate the implication that 10% allowance for excursions would be granted.

Demonstrating compliance with 40 CFR 133.102 as it relates to treatment and performance has been a concern since the implementation of PAA disinfection beginning late 2018. The Stiles Facility compliance point for BOD and TSS treatment and performance has been via a flow weighted composite sample of final effluent collected via two horizontal pipes about 4 feet

below the surface on each side of the disinfection basin. This sampling apparatus dates to the original construction of the treatment plant. To provide continuity of data, it was decided to request a BOD credit calculated on PAA dose and keep everything else the same. The concept of BOD credit appeared to be reasonable based on three studies performed on Stiles Facility effluent by vendors and consultants over the years. However, all were based on grab samples. The Stiles Facility initially used portable Isco autosamplers and later installed a permanent pre-final effluent sampling apparatus (going into service on 9-17-2021) comparable to the final effluent sampling apparatus and embarked upon a formal study, including measuring the increase in soluble BOD across the disinfection tank due to PAA addition. The study revealed additional BOD and TSS removal was occurring across the disinfection tank during the monitoring period (2017-2021) mainly due to additional TSS removal that also lowered BOD. The question now is why select solids are not being captured in the final clarifiers. Getting this issue resolved is very important, especially if the pre-disinfection sampling point is ever used for secondary treatment compliance.

For the Maxson Facility, more than likely, secondary treatment will be based on the existing final effluent sample collected upstream of the disinfection tank and PAA impact on BOD would not be considered.

The third Maxson Facility expansion project was initiated in 2014 at the encouragement of City officials to facilitate expansion of existing industries as well as establishment of new industrial development in the Pidgeon Industrial Park. During the early years of the City of Memphis Pretreatment Program, discharge agreements were signed with industrial users and industries sought to "negotiate" greater discharge limits than currently needed for future expansion. In later years, the City moved to a permit approach which had less perceived opportunities for "negotiation" and began to reign in past promises to industries in an informal effort to avoid over allocation compared to rated treatment capacity. As such, industries began contacting the Mayor's office to secure such commitments outside of the industrial user permit system. For the moment, an informal truce has been reached between existing industries and the wastewater treatment plants with both treatment plants near capacity and able to meet current permit limits. An expansion would allow existing industries to be able to plan for future expansion and allow the City to recruit new industries to the Pidgeon Industrial Park where there is plenty of land and other resources available.

The expansion project initiated in 2014 was broken into five smaller separate GMPs which, taken together, would encompass the desired capacity increase and performance improvements as well as disinfection and influent coarse bar screen replacement. The cost to perform this work was slated to be approximately $251 million with approximately $83 million funding from SRF.

The Maxson Facility upgrades will go a long way toward ensuring compliance in a regulatory scenario that is worst case (i.e. 30-30 limits) especially if rated capacity becomes too large for any one industrial category to qualify for the 10% rated capacity threshold. The Stiles Facility is currently loaded to capacity and should benefit from the secondary treatment adjustment based on its current industrial mix.

On 11-27-2007, the Natural Resources Defense Council petitioned the US EPA to request updated information about secondary treatment nutrient removal capability and to establish new

technology-based nutrient limits as part of the secondary treatment standards. On 3-16-2011, a Nutrient Reduction Framework (NRF) memo was issued to the EPA regional offices to guide EPA technical assistance and collaboration with states. This initiative was subsequently rolled down to the states. On 12-14-2012, the EPA responded to the NRDC petition saying that it agreed that insufficient data exist to draw any general conclusions about the ability of secondary treatment to remove nutrients and that a uniform set of nationally applicable, technology-based nutrient limits was not warranted. In February 2016, the EPA issued the 1st Gulf of Mexico Hypoxia Task Force report. On 10-16-2019, EPA issued the 2nd Gulf of Mexico Hypoxia Task Force report. Both the Maxson and Stiles Facilities were listed as major point source contributors from Tennessee. On 8-17-2020, it was announced that HTF would begin working with SERA-46 and other research groups to evaluate factors that contribute to nutrient loss from agricultural lands, state-level nutrient impairments, and hypoxia in the Gulf of Mexico. In 2021, the HTF announced potential new monitoring sites in the Mississippi-Atchafalaya River basin, particularly to track loads and trends in large rivers to help states evaluate progress toward meeting nutrient reduction goals and to support adaptive management of nutrient reduction strategies. The 4th Gulf of Mexico Hypoxia Task Force report was issued in October 2023. The HTF remains committed to its 2035 goal of reducing the 5-year average areal extent of the hypoxic zone in the Gulf to less than 5,000 square kilometers by 2035, with an interim target for reducing total nitrogen (TN) and total phosphorus (TP) loads by 20 percent by the year 2025.

In addition, the EPA initiated a multi-year, multi-phased national study of nutrient discharges from POTWs to 1) establish a statistically representative, nationwide baseline for nutrient discharge and removal at these facilities, and 2) characterize operational and management practices that result in improved nutrient reduction. A questionnaire was published in the Federal Register on 9-19-2016 but was withdrawn after the presidential election. The City responded to a POTW Nutrient Survey / Voluntary Questionnaire issued by the EPA on 10-22-2019 for both treatment facilities. The EPA compiled the survey responses collected from 2019 through 2021. Survey results to date show more than 1,000 POTWs with different biological treatment types (including both conventional and advanced treatment technologies) can achieve effluent total nitrogen of 8 mg/L and total phosphorus of 1 mg/L. In 2023, the EPA continued to support state programs and making funds available to POTWs through SRF and IRA.

In 2014, TDEC submitted a draft NRF to the EPA for review and subsequently began implementing it. The TDEC NRF prioritizes HUC10 watersheds with 303(d) listed segments. Tennessee is developing a Non-Point Source nutrient reduction strategy for Mississippi River basin (Hydrogeological Unit IV) using a farmer-led approach so it is not known at this time how the Maxson and Stiles Facilities will be affected. The initial state-wide implementation of nutrient reduction would emphasis optimization instead of capital improvement. In 2018, TDEC's emphasis was on developing a larger stake holder group for input on the NRF in general and developing site-specific criteria for three reservoirs in Tennessee as a priority.

At the direction of the governor, in 2019, TDEC and TDA convened the Tennessee Nutrient Strategy Taskforce (the Taskforce), drawing representation from academia, state and local government, wastewater treatment plant operators, the private sector, and non-governmental organizations. In 2021, the Tennessee General Assembly issued joint House Resolution 751 recognizing the work of Tennessee representatives on the Hypoxia Taskforce. Among other objectives, the Taskforce planned to generate a list of municipal wastewater treatment plants in Tennessee with permits for: TN and/or TP limits, Report TN and/or TP or No Reporting

required and, if possible, planned to develop a consolidated spreadsheet with flow, loads and nutrient concentration data for wastewater treatment plants. During 2020-2022, a tool kit was developed by Middle Tennessee State University by EPA grant for use by stake holders. The tool kit is not yet final.

WEF announced on 2-2-2022 a "Nutrient Smart Recognition Program" in collaboration with the EPA. Wastewater treatment facilities would submit on a voluntary basis loading and discharge data for nitrogen and phosphorus as part of a 1-year pilot program. Facilities achieving a 90% removal would receive a Platinum Award as well as lessor awards and awards for innovation.

In 2022, TDEC announced it was starting a new round for its Tennessee Plant Optimization Program (TNPOP) for nutrient reduction. Instead of a state-wide program for nutrients, TDEC indicated it would take matters on a case-by-case basis, including establishing criteria for reservoirs. Federal money would available for states participating on the Gulf Hypoxia Task Force as well. In 2023, the Taskforce continued to meet. A goal was set for the TNPOP to have reached all POTWs by 2030. At the WPC23, TDEC stated that POTWs should use Local Limits to control nutrients, including PFAS. At the WPC24, TDEC stated that it was about to contract with a vendor to implement its TNPOP. The Iowa Nutrient Optimization model has become a common permitting practice across the country. According to Gary Cohen, Nutrient Plant Optimization would possibly be a disincentive to antidegradation.

PFAS has become an increasing concern since the Stiles Facility began land application of biosolids. In March 2024, the USEPA public noticed a nationwide study on PFAS in POTW influent, effluent, and sewage sludge to cover approximately 400 POTWs. The study would collect data on industrial contribution to biosolids PFAS and support the development of effluent limitation guidelines. In April 2024, the USEPA issued the Final Rule for PFAS National Primary Drinking Water regulations identifying 6 specific PFA compounds. In May 2024, the Final Rule was issued designating PFOA and PFOS as CERCLA hazardous substances. In November 2024, the USEPA made a commitment to perform a risk assessment of PFAS in biosolids. In December 2024, the USEPA public noticed an effort to facilitate a review of issued related to PFAS and Biosolids and associated challenges. It is expected that the USEPA guideline for domestic sludge will be 25 ppb and for municipal sludge will be 40 ppb. No questionnaire has been received as of the end of 2024 as originally annou

<u>Corrosion-The Silent Killer</u>

Coupons were installed at the Maxson Facility from April through May 2020 to assess current corrosion rates and project future corrosion conditions. The areas surveyed were rated "Severe" except for the main control room which was rated "Harsh." The Severe designation means that electronic/electrical equipment is not expected to survive. Each area represents a unique challenge especially as the City transitions to SCADA which involves more delicate hardware with sensitive metals designed for modern control room or data center atmospheres not typically encountered at wastewater treatment plant environments. Testing at the Stiles Facility has produced similar results. Corrosion is an equal concern in both POTW and Collection System infrastructure.

Metals

The Pretreatment Program had set allowable limits for many categorical dischargers that resulted in the significant reduction of metals received over the past three decades. With the promulgation of the 503 Sludge Regulations, the various metals entering the plants became a greater regulatory concern. Prior to 1994, the discharge of metals had been controlled and regulated but there were no regulations that defined allowable concentrations of various metals in biosolids. Table 31 shows the metals concentrations over the years in the plants' influent and biosolids. The tables utilize the combination of the six most common metals, zinc, copper, nickel, chromium, lead, and cadmium. As the table shows, influent metals loadings have generally decreased over the past 4 decades. However, this past approach used to evaluate metals impact may not be entirely appropriate when the concern is specific metals, including molybdenum, nickel, and zinc. In addition, based on the 2023-24 metals investigation at both plants, metals can selectively accumulate at the treatment plants over time, impacting both activated sludge health and biosolids disposal as thresholds of concern are gradually approached.

Table 31 also shows the average limit that would apply to the Maxson and Stiles Facilities either as land application or surface disposal under the 503 regulations as well as the metals concentrations for biosolids sampled in 2024. The metals are arsenic, cadmium, chromium, copper, lead, mercury, molybdenum, nickel, selenium, and zinc. At the Stiles Facility, influent metals decreased from 1985 to 2018 and have now been increasing. Metal concentrations in surface-disposed bio-solids generally decreased from 1980 to 2021 and now are increasing, lagging influent metals by 3 years. As stated, there is a growing concern with increasing levels of specific metals at either treatment plant as well as bioaccumulation.

With the introduction of land application of biosolids in 2018 for the Stiles Facility, the frequency of sampling of biosolids for metals has increased above the bare minimum requirements listed in the NPDES permits indicating seasonal variation, multi-year trends, and possibly episodic increased contribution of metals by industrial users previously unnoticed. Table 32 and Table 33 show monthly metals results for the Maxson and Stiles Facilities, respectively.

More frequent influent sampling for metals as well as specific interceptor sampling (i.e. Nonconnah Interceptor) is warranted as a proactive measure to evaluate any trends, confirm existing data, and identify potential specific sources based on sampling, especially related to Molybdenum which has a lower threshold, 75 mg/kg, for regulatory concern at the Maxson Facility. Figure 34 shows monthly molybdenum concentration trends in Maxson Facility biosolids since 2018. Nickel and Zinc are also a concern, especially at the Stiles Facility and as evidenced by the process upset that occurred at the Stiles Facility beginning late 2023 running into 2024 which was attributed, all or in part, to elevated levels of Nickel and/or Zinc in the mixed liquor suspended solids. Figures 35 and 36 show monthly nickel and zinc concentrations in Stiles Facility biosolids since 2018.

Personnel

Previously, all but one of the classifications represented by 369D feature apprenticeship-like step raises based on time in position and financial incentives for certifications. Except for craft unions, many non-exempt employees at the treatment plants came in at the lowest classification and promoted based on most qualified applicant.

In 2011, 369D negotiated a clause in the MOU stating the permanent filling of any vacancy be promotional only to the senior most qualified applicant, whenever possible, except those classified as entry level positions as designated by the Director of Public Works and replaced incentive pay with automatic promotion for maintenance mechanics from M1 to M2. As explained later, this seemingly curtailed management's ability to hire external candidates for the M2 position. Public Works hired a consultant in 2011 to perform a job analysis showing that the Mechanical Operator was not entry level for either the wastewater treatment operator or the maintenance mechanic career paths.

In 2013, Public Works created its own apprenticeship-like program conforming with the state wastewater treatment operator certification requirements. The program consisted of non-union trainee position with a 36-month probationary status that would act as an entry point to the regular wastewater treatment operator position. The obtainment of a Grade 4 certification was the condition for continued City employment as a regular wastewater treatment operator.

The City has elected in the past not to participate in joint apprenticeship and training committees as defined by the U.S. Department of Labor or contributing to any union training funds. The recruitment of maintenance mechanics is hampered by lack of applicants that meet minimum qualifications for journey level or graduate from vocational schools (i.e. pre-apprenticeship program) and the inability to recruit more experienced mechanics with specialty credentials at higher pay due to union restrictions (i.e. Maintenance Mech 2 is promotion-only due to union agreement).

The Workforce Innovation and Opportunity Act (WIOA), signed into law on July 22, 2014, promotes the use of Registered Apprenticeship programs. The State of Tennessee created Apprenticeship Tennessee in October 2020 to work with employers and local educational institutions to develop and expand the use of federally registered apprenticeship programs in the state. Public Works has the option to participate in a registered apprenticeship program, develop a registered program of its own, contribute to a union training fund, or create its own apprentice-like program conforming to its own unique requirements.

In September 2021, HR approved a revised salary progression plan to allow wastewater treatment trainees to advance to WTO1 positions after obtaining Grade 3 licenses. This lowered the minimum qualifications for trainees which resulted in trainees at wider range of abilities and higher turnover.

In 2022, the step raises were eliminated from the 369D positions as part of labor negotiations allowing management to bring in external candidates or promote internal candidates at full top-out pay. In addition, the 2022 MOU required that Public Works utilize the Maintenance Helper position to hire employees that could later apply for M1 positions.

The year 2024 began with approximately 10 wastewater treatment trainees and ended with net 21 trainees. 2024 began with 20 grade 4 license holders and ended with net 20 grade 4 license holders. During the year, one Grade 4 license was obtained and one Grade 4 license lost.

The M1 Trainee program was initiated in 2024 after several months of development. This program is 48-months allowing the trainee to attend Moore Tech and obtain an applied

technology degree in advanced welding while also working full time at the treatment plant. A total of 5 M1 Trainees were hired during 2024 with plans to hire more in 2025.

Covid 19 Testing in Wastewater

The treatment plants submitted raw wastewater samples to two different labs throughout the year as part of a national initiative to correlate Covid19 concentration with number of human cases at no cost to the City. In 2023, the Biobot program transitioned to the State of Tennessee for both treatment plants. The Stiles Facility began participating in the Wastewater SCAN program beginning about July 2023 receiving a monthly stipend for its efforts. The testing included SARS, RSV, Influenza, Monkey Pox, and Hepatitis among others. The treatment plants continued participating in both the State of Tennessee and SCAN programs.

COD Surcharge versus Disinfection Costs

The COD Surcharge is expected to cover 50% of disinfection costs. In 2024, total COD revenue is estimated at $11,563,573 versus $24,609,321 in disinfection cost. The COD Surcharge covered 47% of disinfection costs. A comparison of COD Surcharge Revenue versus Disinfection costs over past years is provided below:

Year	TEM Surcharge	MCS Surcharge	TEM Cost	MCS Cost	Total Surcharge	Total Cost	Total Percent Cost Covered
2019	$8,333,534	$4,537,792	$0	$14,075,692	$12,871,326	$14,075,692	91%
2020	$7,900,285	$3,186,495	$0	$10,519,858	$11,086,780	$10,519,858	105%
2021	$11,057,984	$5,034,881	$4,145,504	$14,243,752	$16,092,865	$18,389,256	88%
2022	$9,227,440	$4,808,020	$8,928,167	$13,339,685	$14,035,460	$22,267,852	63%
2023	$6,589,852	$4,494,523	$6,691,536	$15,048,244	$11,084,375	$21,739,780	51%
2024	$7,312,435	$4,251,138	$7,511,815	$16,894,680	$11,563,573	$24,406,495	47%

The following table shows the percent of disinfection cost covered at each treatment plant related COD Surcharge from industries served by the treatment plant:

Year	TEM Cost Covered	MSC Cost Covered
2019		32%
2020		30%
2021	121%	35%
2022	54%	36%
2023	67%	30%
2024	57%	25%

As shown above, the COD Surcharge revenue from Maxson Facility industrial users, subsidizes the Stiles Facility disinfection costs.

Sewer Fees

The sewer fee history is summarized below:

Date	Volumetric	BOD Surcharge (250 mg/l)	TSS Surcharge (300 mg/l)	COD Surcharge (800 mg/l)	Ordinance
5-15-1979	28.52 cents	1.7 cents	0.9 cents	-	#2877
7-1-1981	-	3 cents	5 cents	-	#3127
1-1-1982	58.68 cents	2.7 cents	4.6 cents	-	-
7-1-2004	87.5 cents	3.65 cents	5.96 cents	-	#5036
7-1-2008	98.5 cents	3.86 cents	6.21 cents	-	#5239
7-1-2009	105.3 cents	-	-	-	#5312
8-1-2009	-	3.88 cents	6.61 cents	-	-
7-1-2010	226.7 cents	4.28 cents	7.16 cents	-	#5356
1-1-2018	287 cents	-	-	-	#5651
1-1-2020	332 cents	-	-	3.96 cents	#5651
1-1-2021	-	-	-	4.821 cents	#5651
1-1-2022	-	-	-	4.821 cents	#5651

The Maxson Facility rated capacity is 90 mgd for flow, 490,000 ppd for influent BOD and 410,000 ppd for influent TSS. The Stiles Facility rated capacity is 135 mgd for flow, 275,000 ppd for influent BOD and 374,000 ppd for influent TSS. Influent flow and loading trends are shown in Figures 9-11 for the Maxson Facility and in Figures 12-14 for the Stiles Facility. For the Maxson Facility, Figure 9 shows annual average daily flow and average daily dry weather flow since 1976. Figure 10 shows annual average influent and industrial BOD loading since 1976 and 1989 respectively. Figure 11 shows annual average influent and industrial TSS loading since 1976 and 1989 respectively. For the Stiles Facility, Figure 12 shows annual average daily flow and average daily dry weather flow since 1978. Figure 13 shows annual average influent and industrial BOD loading since 1978 and 1992 respectively. Figure 14 annual average influent and industrial TSS loading since 1978 and 1992 respectively.

Since 1979, the Additional Treatment Cost charged to industries has increased at an annual rate of 2.09% for BOD and 4.76% for TSS and the volumetric charge for all users has increased at an annual rate of 5.65%.

The COD surcharge was intended to cover 50% of disinfection cost. Below is a summary of annual estimated COD surcharge revenue and disinfection cost:

Year	Approximate COD Surcharge Revenue	Disinfection Cost	Percentage covered by COD Surcharge
2019	$11,067,792	$14,075,692	85%
2020	$11,088,780	$11,519,858	107%
2021	$16,092,865	$18,389,255	98%
2022	$14,035,460	$22,267,852	64%
2023	$11,522,083	$21,739,780	53%
2024	$11,563,573	$24,406,495	47%

FIGURES

TABLES